Car-Following Models: A Multidisciplinary Review

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***Abstract*—Car-following (CF) algorithms are crucial components of traffic simulations and have been integrated into many production vehicles equipped with Advanced Driving Assistance Systems (ADAS). Insights from the model of car-following behavior help researchers to understand the causes of various macro phenomena that arise from interactions between pairs of vehicles. Car-following Models encompass multiple disciplines, including traffic engineering, physics, dynamic system control, cognitive science, machine learning, deep learning, and reinforcement learning. This paper presents an extensive survey highlighting the differences, complementarities, and overlaps among microscopic traffic flow and control models based on their underlying principles and design logic. It reviews a range of representative algorithms, from theory-based Kinematic Models, Psycho-Physical Models, and Adaptive Cruise Control Models to learning-based algorithms like Reinforcement Learning (RL) and Imitation Learning (IL). Additionally, it considers the impact of large Generative AI (GenAI) models, categorized as Knowledge-Driven models. The survey discusses the strengths and limitations of these models, explores their applications in various contexts, and summarizes available datasets across different domains to address knowledge gaps. By synthesizing historical developments and current trends, this survey provides a comprehensive overview of the evolution and future directions of car-following models. It highlights the importance of interdisciplinary approaches and the need for continued innovation to address emerging challenges.**

***Index Terms*—Car-Following Behavior, Adaptive Cruise Control, Theory-based, Learning-based, Knowledge-driven.**

I. Introduction

To furnish a complete picture of traffic flow, vehicles on the road are studied as a whole - not merely as isolated particles. Car-following (CF) is the fundamental driving behavior that forcefully constrains the ego-vehicle from adopting a specific pattern in the presence of other vehicles (Figure 1). Since the 1950s or even earlier, many researchers have focused on driver behavior modeling and applied such models to the simulation of Human Driver Vehicles (HDVs) to investigate micro- and macro-traffic phenomena [1], [2]. The expansion of autonomy and connectivity in the transportation system has escalated the hype for integration of car following models with Connected and Automated Vehicle (CAV) technologies, such as collaborative driving, and Vehicle-to-Everything (V2X) communication. Autonomous Vehicles (AVs) with different parameter settings or objective functions become a new source of variety and add more complexities to the existing roadway systems. The heterogeneity of vehicle types and driver behaviors gives rise to various challenges for traffic operation, which must be comprehensively analyzed before upgrading existing infrastructure. Car-following models become a powerful tool to address challenges mentioned above and answer those critical questions in the upcoming era of the connected and automated transportation systems. By analyzing the interaction between vehicles, researchers can identify efficient driving strategies that includes optimizing speeds, minimizing unnecessary acceleration and deceleration events, and maintaining optimal following distances to ensure smooth traffic flow. Modern adaptive cruise control (ACC) systems already use basic car-following principles to maintain safe distances between vehicles.

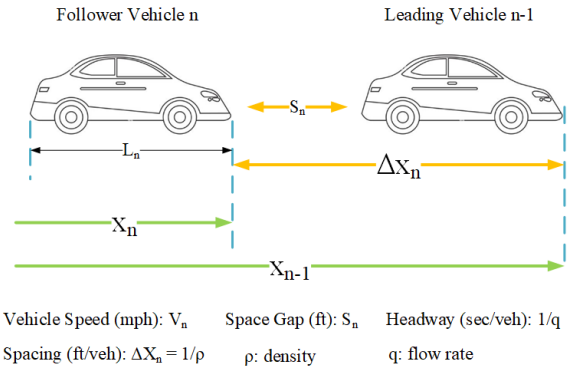

Fig. 1. Car-Following and Microscopic Traffic Flow Properties.

Existing survey papers [3]–[19] lack of structured framework and often have limitations in terms of scope and perspective (see Table I). For example, a frequently referenced paper [4] mainly surveyed from traffic psychologic perspective and did not consider Intelligent Driver Model (IDM) and Optimal Velocity Model (OVM). Some surveys [9], [11] reviewed multiscale traffic flow models from the engineering and physics perspective, while many control theory models were omitted. Other literature reviews [8], [10] are limited to theory-based models and inadequate to reflect AI/ML models. By taking into account various domains and angles, this review paper aims to provide a more structured categorization of the car following models (see Figure 2), addressing the potential gaps left by previous surveys with a narrower focus. A multidisciplinary review can be particularly beneficial in fields like intelligent vehicle technology, where insights from different disciplines – such as electrical & computer engineering, mechanic engineering, traffic flow theory, physics, and optimal control – are crucial for developing a complete picture of how these agents operate and interact with their environment. This review paper not only integrates diverse knowledge but also promotes the discovery of innovative solutions that might be overlooked when considering a single perspective. Our

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contributions can be itemized as follows:

- **Multidisciplinary Perspective**: Unlike previous surveys that focus on a single aspect of car-following models (e.g., kinematic characteristics, specific applications, or human factors), our paper provides a comprehensive review from multiple disciplinary perspectives. This broader approach allows for a more thorough evaluation of the state-of-the-art, challenges, and future directions of car-following models, making a unique contribution to the field.
- **Novel Taxonomy for Car-Following Models**: We introduce an effective taxonomy to categorize the broad spectrum of car-following model research. Our taxonomy is not just a listing; it delves into the underlying design principles, offering a structured framework that surpasses the categorizations found in previous surveys. This novel approach allows for a more intuitive understanding of the field's evolution and current state.
- **Extensive Coverage of Application Cases**: Our survey extends the discussion to various application cases built upon car-following models, highlighting their significant impact on the environment, safety, mobility, and operation efficiency. This comprehensive review of applications is unprecedented in existing literature, providing valuable insights into the practical implications of car-following models.
- **Future Directions with Human-like Reasoning**: We underscore the future direction of integrating AI-driven methods with theory-based approaches to embed human-like reasoning processes into car-following models. This perspective is critical for developing more advanced, realistic models that can better mimic human driving behaviors, addressing a gap in current research trends.

By providing a thorough evaluation of current challenges and paving the way for future research directions in car-following models, our review marks a distinctive and valuable contribution to the literature.

II. Theory-based Models

Theory-based models are easy to interpret and highly flexible, which can be adapted to various applications. They are also easy to perform stability analysis with traditional control techniques. Kinematic Models emphasize the physical aspects of vehicle motion and dynamics; Psycho-Physical Models integrate human factors, particularly how drivers perceive surrounding environment and their decision-making processes; Adaptive Cruise Control (ACC) Models emphasize the mechanistic low-level control to maintain safe and efficient vehicle dynamics.

A. Kinematic Models

This principled car-following models are based on kinematic relationship, considering factors as velocity, headway and distance. First-order models are simpler models that prescribe the motion of a vehicle in terms of its velocity. Second-order models take acceleration into account, which can provide a more detailed simulation of real-world driving behavior. Despite their varied mathematical frameworks, many of these models share a common heuristic basis, making them interchangeable under specific scenarios. For instance, the OVM combined with Euler updates for position and speed is mathematically equivalent to Newell's model when Newell's reaction time parameter is set to the inverse sensitivity parameter (a^{-1}) of the OVM [9]. Moreover, Gipps' model and the IDM model share a set of parameters derived from the same driving strategy that drivers tend to keep a safe distance from the leading vehicle with comfortable speed or accelerations. Table II provides a comparison among kinematic models.

1) First-Order Kinematic Models:

The first-order kinematic car-following model, an early approach in car-following modeling, is a mathematical description of the motion using the classical law of motion to update the vehicle's position at each time step. This type of model often features parameters that possess meaningful physical interpretations, and these principled models treat each driver-vehicle unit as a particle or automaton. As an elementary model, Herrey and Herrey [20] introduced the concept of influence space, considering minimum safety distance, and validated their hypothesis using time-space diagrams derived from field data. Pipes [21] describes a linear relationship between the vehicle speed with respect to the minimum safe headway suggested by the California Motor Vehicle Code. Kometani and Sasaki performed stability analysis by introducing a transfer lag to Pipes' traffic dynamic model [22] and built a safety index equation that is measured to represent the safety level in sinusoidal traffic flow [23], [24].

Gipps' Model Gipps [25] developed a more advanced version of these explicit physics-based car-following models, assuming that a driving agent adapts the speed to smoothly and safely proceed behind its leader. Gipps model puts two constraints on the CF behavior: the vehicle will not exceed the driver's desired speed or a vehicle-specific free-flow speed, and the following driver selects a speed that can bring the follower vehicle to a safe stop. In addition, while the preceding vehicle decelerates, a safety margin should be added to the following driver's reaction time, limiting the speed of the following vehicle based on the performance of the leader vehicle, particularly with respect to its deceleration capability. The formulation is shown as follows:

$$v_n(t+\tau) = \min\left\{ v_n(t) + 2.5a_n\tau\left(1-\frac{v_n(t)}{V_n}\right)\left(0.025+\frac{v_n(t)}{V_n}\right)^{1/2}, \right. \\ \left. b_n\tau + \sqrt{b_n^2\tau^2 - b_n\left[2\left[x_{n-1}(t) - s_{n-1} - x_n(t)\right] - v_n(t)\tau - \frac{v_{n-1}(t)^2}{\hat{b}}\right]} \right\} \tag{1}$$

In the formulation, a_n is the maximum acceleration; b_n is the most severe braking that the driver of vehicle n wishes to undertake; v_n is the speed at which the driver of vehicle n wishes to travel; V_n is the desired speed of vehicle n; τ is the reaction time; $\hat{b}$ is the estimate of b_{n-1}. Because of its explicit physical meaning, modified Gipps' model was used in several traffic simulation software [26], [27]. Krauss's model [28] and Gipps' model are categorized within the same "family" of CF models, because both models fundamentally consider braking distances to derive interactions between vehicles. This shared focus on braking distances and the resultant dynamics

TABLE I
SUMMARY OF CAR-FOLLOWING MODEL SURVEY PAPERS

Paper Title	Year	Scope of Work	Limitations
Car following models [3]	1992	Summarized car-following models using mathematical model descriptions, stability analysis, calibration from empirical testing and evaluation and finally automatic vehicular control systems.	It only covers stimulus-response models from psychological and behavior perspective.
Car-following: a historical review [4]	1999	Summarized theoretical early-stage car-following models from psychological and behavior perspective, such as safety distance, desired gaps, stimulus-response process.	It doesn't include some important models such as Intelligent Driver Model (IDM) and Optical Velocity Model (OVM).
Comparison of Car-following models [5]	2004	Compared theoretical car-following models used in well-known traffic micro-simulation software, including AIMSUN, MITSIM, Paramics and VISSIM.	From a software perspective to evaluate how different programs could replicate real world scenarios.
Driving Behavior: models and challenges [6]	2007	Reviewed and expanded review lists in [4] with IDM, OVM and Cellular Automata. Discussed the challenges associated with the myopic driver behavior models.	Only listed theory-based car-following models from human behavior perspective.
Car-following models fifty years of linear stability analysis a mathematical perspective [7]	2011	A comprehensive survey of linear stability types, illustrated with straightforward examples and rigorous mathematical criteria.	From a control theory perspective, it introduced a general framework that simplifies the analysis of linear stability by requiring only three partial derivatives.
Microscopic car-following model for the traffic flow: the state of the art [8]	2012	Summarized and reviewed theory-based car-following models, also provided the stability analysis from control theory point of view.	Only reviewed theory-based car-following models from human behavior perspective.
Traffic flow dynamics [9]	2013	Categorized existing models as elementary models and strategic models by design principles in microscopic traffic flow modeling.	Only covers kinematic models from the physics perspectives.
Incorporating human-factors in car-following models: a review of recent developments and research needs [10]	2014	Introduced various human factors that could potentially impact driving styles.	Only summarized theory-based car-following models from engineering and human behavior perspective.
Genealogy of traffic flow models [11]	2015	Summarized macro-, meso- and micro- traffic flow models, offered an historical model tree by tracing genealogy of the models.	Reviewed and surveyed car following models from the traffic engineering and physical perspectives
A state-of-the-art review of car-following models with particular considerations of heavy vehicles [12]	2015	Categorized CF models into classical models and AI models. Emphasized the impact of heavy trucks in car following behavior.	Under AI category, only fuzzy logic models and Neural Network models were considered. Under the classical model category, it doesn't include most recent ACC models
Simulation Strategies for Mixed Traffic Conditions: A Review of Car-Following Models and Simulation Frameworks [13]	2020	This review paper focus on Heterogeneous vehicle types and mixed traffic flow modeling	Inherit the same categorization from previous review paper [4] and mainly from simulation software perspective
A review of car-following models and modeling tools for human and autonomous-ready driving behaviors in micro-simulation [14]	2021	Emphasized the differences between human-driven vehicles and autonomous vehicles in car-following models.	Inherit the same categorization from previous review paper [4] and mainly from simulation software perspective.
The Car-Following Model and Its Applications in the V2X Environment: A Historical Review [15]	2021	Emphasize applications of car-following models in the V2X environment	Use the similar categorization framework from previous review paper [4], [6] to categorize car-following models
Modeling the Car-Following Behavior with Consideration of Driver, Vehicle, and Environment Factors: A Historical Review [16]	2022	This review paper surveyed car models that consider diverse driver styles, physical and dynamic characteristics of vehicles, as well as environmental factors.	Inherit the similar categorization from previous review paper [4], [6], [15] and is developed from traffic engineering perspective.
Modelling and simulation of (connected) autonomous vehicles longitudinal driving behavior: A state-of-the-art [17]	2023	This paper reviewed parameters for (C)AVs modelling. It also summarizes the lately published data-driven models for (C)AVs CF behavior, and identification of KPIs used for impact assessments.	Mainly focus on CF models' parameters that are more crucial and sensitive in differentiating (C)AVs from human driven vehicles.
Microscopic Modelling of Car-Following Behavior: Developments and Future Directions [18]	2023	This paper analyzes advantages and limitations of four main types of CF models: Kinematics-based, dynamics-based, psychological-based, and learning-based.	Didn't provide subcategories and missed Cellular Automata, and ACC models.
Car-following models for human-driven vehicles and autonomous vehicles: A systematic review [19]	2023	Reviewed CF models in terms of accuracy, calibrations, stability analysis and specific model parameters for HDVs and AVs simulation	Limited to Traffic Engineering perspective for categorization and applicable background

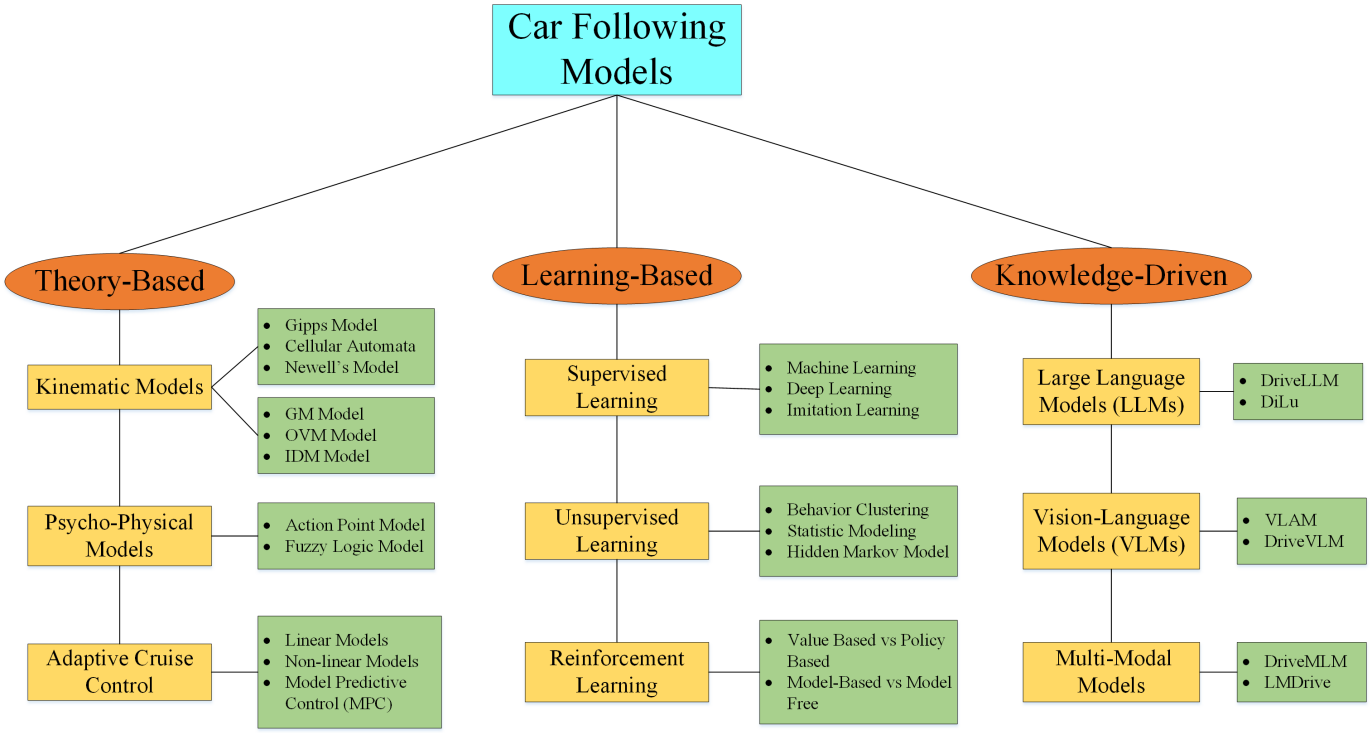

Fig. 2. Proposed Taxonomy of Car Following Models (CFMs) and Representative Algorithms. Kinematic Models are divided into first-order and second-order subcategories. Knowledge-driven Large GenAI Models are a revolutionary category, providing new insights and solutions with commonsense.

establishes a conceptual link between the two models. Rakha and Wang [29] developed an analytic calibration steps of Gipps model for steady state behavior. Some studies [30] showed that the Gipps' model also allows reproducing the Phantom Traffic Jam, although the model is formulated as a first-order category. A modified Gipps model [31] addressed the potential collision when the most severe braking the following vehicle wishes to undertake is harder than the perceived deceleration rate of lead vehicle.

Newell's Model In Newell's Model [32], the time-space trajectory of the n^{th} vehicle is essentially the same as the $(n-1)^{th}$ vehicle, except for a translation in space and in time. This is a very simple rule for car following behavior compared with other models; however, it is often much more accurate than many. Newell's Model is defined as:

$$x_n\left(t+\tau_n\right)=x_{n-1}\left(t\right)-d_n \tag{2}$$

where $x_n(t)$ is a piecewise linear trajectory of vehicle n, which will be a simple translation of piecewise linear $x_{n-1}(t)$ by a distance d_n and a time lag τ_n. By translating their trajectory forward or backward, drivers can manage the spacing between their vehicle and the preceding vehicle, ensuring a suitable and desirable headway. Newell's Model implies that drivers have the ability to adapt their position relative to the preceding vehicle by adjusting their velocity and position in a controlled manner, thereby maintaining a safe and comfortable driving experience. These fixed translations have been validated with video-taped trajectory data and described by a bivariate normal distribution [33]. L-L model [34] introduced a behavior deviation term to the basic Newell's model and find that timid and aggressive drivers are the cause of spontaneous traffic jam. Later on, an asymmetric behavioral model (ABM) is developed by adding five parameters per driver to capture different driver reactions to traffic oscillations [35]. To account for randomness in driver behaviors and further understand traffic oscillations explained by string instability, stochastic components were added to the steady-state Newell's equations [36], [37]. Newell's model can replicate the formation and propagation of traffic waves with reasonable accuracy, and is particularly effective in large-scale network simulations [39]. By considering vehicle/engine dynamics and driving situation-dependent stochasticity, multidimensional stochastic Newell car-following model (MSNCM) is proposed for emission estimation [40].

Cellular Automata Cellular automata models were first introduced in the study [41], defined by a one-dimensional string of cells with two states (occupied and empty). The cell updating policy is known as Rule 184, described as:

$$v_\alpha\left(t+1\right)=\begin{cases}1, & \text{if } x_{\alpha-1}\left(t\right)-x_\alpha\left(t\right)>1\\ 0, & \text{otherwise}\end{cases} \tag{3}$$

$$x_\alpha\left(t+1\right)=x_\alpha\left(t\right)+v_\alpha(t+1) \tag{4}$$

where α is the vehicle index. x_α is the location of the vehicle α, v_α is the speed of vehicle α.

A popular and simple Cellular Automata (CA) model is Nagel-Schreckenberg Model (NSM) [42]. The NSM model can only update speed and acceleration on multiples of $7.5m/s$ and $7.5m/s^2$ due to the coarse discretization of space and time, and one cell corresponds to effective vehicle length and

one-time step under car-following mode. First modification was made by adding a slow-to-start rule to the set of NaSch rules [43] Some refined CA models include the Velocity Dependent Randomization (VDR) Model [44], Benjamin-Johnson-Hui (BJH) Model [45] and Anticipation Model [46], which generalized slow-to-start rule in Takayasu-Takayasu (TT) model. The KKW (Kerner–Klenov–Wolf) [47] is a CA model within the framework of three-phase traffic theory and allows for smaller interval states. While the critical aspect of the KKW model allows the model to simulate the variability and unpredictability of real-world driving behavior, it can also lead to instability in the simulation. Bham and Benekohal (2004) [48] proposed the CELLSIM model to incorporate relative speed and distance headway into CA framework. Chakroborty and Maurya (2008) [49] conducted comparable analysis of different CA based traffic flow models in terms of microscopic properties (like time headway distribution, acceleration noise, stability, etc.). By focusing on spatio-temporal patterns, phase transitions, and growth patterns of oscillations, cellular automata can capture the intricate behaviors observed in real-world traffic, such as the formation and dissipation of traffic jams, and the transition from free-flow to congested traffic conditions [50], [51]. CA based simulation models are often used for heterogeneous traffic flow analysis, including pedestrian and bicycle flow dynamics after adjusting their parameters to reflect the lower speeds and smaller sizes [52]–[59].

2) Second-Order Kinematic Models:

Second-order kinematic models are based on the concept that the following vehicle adjusts acceleration (or deceleration) in response to the observational motion of the leading vehicle to maintain a desired speed and safety distance. The second-order kinematic models often incorporate sensitivity factors or driver preference parameters to capture the driver's response to various stimuli or conditions in the traffic environment. These sensitivity factors represent the driver's characteristics, driving styles, and preferences, which are widely used for analyzing factors that might affect the car-following process by formulating in different ways, such as adding stochastic factors from additional stimuli and variations in sensitivity parameters.

General Motors Model General Motors (GM) model family is also known as GHR (Gazis-Herman-Rothery) model, which was first developed by Chandler et al. [60] and Herman et al. [61] at the General Motors research laboratory. GM model family is based on the stimulus-response process of humans, and they have evolved through five generations. The stimuli term can cause the response to be acceleration, deceleration or constant speed. The model formulation is shown as follows:

$$\frac{dv_n(t+\Delta)}{dt} = \lambda[v_{n-1}(t) - v_n(t)] \tag{5}$$

λ denotes the sensitivity of the driver, Δ denotes the response time of the driver. Gazis et al. [62], [63] found that the sensitivity λ was inversely proportional to the space headway. It was found that different data and different experiments offered a different combination of m and l with the same car following model [64]–[66], which blocks the application of the GM models. Variations of GM Model are employed to analyze driver behaviors, for instance, a "multi-following model" [67] where drivers consider several vehicles ahead. Researchers applied GM models to emphasize the asymmetric nature of car following behavior for distinct treatment of acceleration and deceleration phases [68].

Optimal Velocity Model Optimal Velocity Models (OVMs) are a set of models based on the assumption that each vehicle pursues a legal velocity. Bando et al. [69], [70] proposed the first Optimal Velocity (OV) Model that each driver controls acceleration or deceleration to achieve an optimal velocity $V(\Delta x_n)$ according to headway distance Δx. The original formulation is expressed as follows:

$$\frac{dv_n}{dt} = a[V\left(\Delta x_n(t)\right) - v_n(t)] \tag{6}$$

The optimal velocity function is as below:

$$V\left(\Delta x_n\left(t\ \right)\right) = \frac{v_{max}\left[\tanh\left(\Delta x_n\left(t\right) - \ h_c\right) + \tanh\left(h_c\right)\right]}{2} \tag{7}$$

where a is the sensitivity coefficient, v_{max} is the maximum velocity. h_c is the safety distance. $\Delta x_n(t)$ is the relative distance to the preceding vehicle, and $v_n(t)$ is the velocity of the following vehicle.

OVM has lead to many variations, demonstrating its interpretability and adaptability to different aspects of traffic flow and vehicle dynamics. For instances, General Force Model [71] introduced terms that represent the attractive force and repulsive force; Full Velocity Difference Model [72] emphasizes the relative velocity between a following vehicle and the vehicle directly in front of it; Coupled Map CF model [73] is built on OVM to describe the interactions between vehicles with the concept of coupled map lattices. By considering Multi-Vehicle interaction, researchers expanded OVM to consider both preceding or following vehicles [74]–[76]. By incorporating macroscopic and microscopic models, Nagatani [77] proposed a lattice hydrodynamic model combining macroscopic hydrodynamics with microscopic optimal velocity; Chiarello et al. [78] used OVM to examine the follower-the-leader interactions in the transition from the inhomogeneous Aw–Rascle–Zhang (ARZ) model to the Lighthill–Whitham–Richards (LWR) model; By considering the asymetric and lane separation, Gong et al. [79] developed an asymmetric full-velocity difference, car-following model. Jin et al. [80] considered the lateral separation characteristics between the followers and leaders on a single-lane highway based on the full optimal velocity model. A confined Full Velocity Difference (c-FVD) model [81] was proposed that limits the acceleration or deceleration generated by the existing FVD models to a reasonable level. Each variation or extension addresses specific challenges or aspects of real-world driving, contributing to the broader understanding of traffic behavior. Shang et al. (2022) proposed an asymmetric optimal velocity relative velocity model (AOVRV), considering common velocity variation [82].

Intelligent Driver Model (IDM) The Intelligent Driver Model (IDM) was proposed by Treiber et al. [83], which is also known as a strategy-based model with a list of basic

TABLE II
SUMMARY OF REPRESENTATIVE THEORY BASED CAR FOLLOWING MODELS FROM KINEMATIC PERSPECTIVE

Model	Authors & Year	Key Strengths	Key Limitations
Gipps Model	Gipps (1981) [25] Krauss (1998) [28] Shah et al. (2023) [31]	1. Safety considerations: Gipps Model considers braking capabilities and adding a safety margin, even if the leading vehicle suddenly starts to slow down at this maximum rate. 2. Explicit Parametrization: Gipps Model describes various aspects of driver behavior and vehicle performance, including maximum acceleration, comfortable deceleration, desired speed, and reaction times. These parameters can be adjusted to simulate different driving styles, vehicle types, and traffic conditions.	1. Deceleration Assumption: The braking action a driver decides to take is based on ideal scenario that assumes the driver is always able to accurately judge the necessary braking effort. Different combinations of b_n and $\hat{b}$ could lead to potential collision [31] under extreme scenarios.
Newell's Model	Newell (2002) [32]; Laval and Leclercq (2010) [34]; D. Chen et al. (2012) [35]; Laval et al. (2014) [36]; Xu and Laval (2020) [37]; Meng et al. (2021) [40];	1. Simplicity: Newell's Model is easy to implement and analyze. 2. Traffic wave simulation: Newell's Model is often used to simulate traffic waves caused by traffic signals or other disturbances in the interrupted traffic flow.	1. Limited Representations: Newell's Model simplifies the driving process without any driver-related parameters, which may lead to less realistic representations.
Cellular Automata	Wolfram (1992) [41] Takayasu and Takayasu (1993) [43] Benjamin et al. (1996) [45] Barlovic et al. (1998) [44] Knospe et al. (2000) [46] Kerner et al. (2002) [47] Bham and Bebekohal (2004) [48]	1. Scalability: CA models are computationally efficient as they operate on a discrete grid with simple updating rules, making them suitable for large-scale traffic simulations. 2. Flexibility: CA models are versatile in representing various traffic scenarios, such as heterogeneous traffic components and meta-stability.	1. Purely Particle-Based: CA models treat vehicles as particles moving on a grid without considering their actual physical dimensions beyond the grid size. This abstraction can overlook some aspects of vehicle behavior and interactions that are influenced by the physical size and the mechanics of vehicles, for instance, the distance headways.
GM Models (Gazis-Herman-Rothery)	Chandler et al. (1958) [60] Herman et al. (1959) [61] Gazis et al. (1959) [62] Gazis et al. (1961) [63]	1. Stimulus-response foundation: The GM model provides generalizable analytic framework from the stimulus-response perspective. $Response = f(sensitivity, stimulus)$. It also allows customization based on different driver characteristics.	1. Deterministic Stimulus: GM model assumes precise perception of stimulus, ignoring the inherent vagueness of CF process. 2. Parameter Calibration: Calibrations and Validations of GM model's parameters are often contradictory, as CF behaviors vary with traffic and flow conditions.
Optimal Velocity Model (OVM)	Bando et al. (1995) [69] Nagatani (1998) [77] Helbing and Tilch (1998) [71] Konishi et al. (1999) [73] Lenz et al. (1999) [74] Nakayama et al. (2001) [75] Jiang, Wu and Zhu (2001) [72] Hasebe et al. (2003) [76] Gong et al. (2008) [79] Jin et al. (2010) [80] Yu et al. (2019) [81] Shang et al. (2023) [82]	1. Interpretability: The OVM is straightforward with few parameters, aligning well with real-world driving behavior. 2. Flexibility: The OVM can be easily extended or modified to include additional factors, e.g. macroscopic traffic conditions, safety constraints.	1. Simplifications: The assumption that drivers always pursue an optimal velocity as mental intention may not be true in all real-world situations.
Intelligent Driver Model (IDM)	Treiber et al. (2000) [83] Treiber and Helbing (2003) [84] Treiber et al. (2006) [85] Kesting, Treiber and Helbing (2010) [86] Derbel et al. (2013) [88] Eggert et al. (2015) [90] Hoermann et al. (2017) [92] Treiber and Kesting (2017) [93] Hubmann et al. (2018) [95] Zhou et al. (2024) [98]	1. Smoothness and safety: IDM captures key strategies of driving behaviors, such as maintaining a safe following distance, smooth acceleration, and comfortable deceleration. 2. Integral and Versatility: The IDM integrates the follower's speed, relative velocity, and distance headway, offering the versatility to apply in both uninterrupted and interrupted flows across highways and urban settings.	1. Coefficient's Meaning: Certain parameters (e.g., Acceleration Exponent δ) don't contain physical meaning.

assumptions. For IDM models, the vehicle's acceleration is described as below:

$$\frac{d}{dt}v = a_{max}\left(1 - \left(\frac{v\left(t\right)}{v_{desired}}\right)^{\delta} - \left(\frac{s^{*}\left(v, \Delta v\right)}{s}\right)^{2}\right) \quad (8)$$

where a_{max} is the maximum acceleration of the vehicle. v is the current velocity of the vehicle. $v_{desired}$ is the desired velocity in free traffic. δ is the acceleration exponent (usually set to 4). s is the current bumper-to-bumper gap. Δv is the relative velocity (velocity difference) with respect to the vehicle ahead.

The desired minimum gap s^{*} is given by:

$$s^{*}\left(v, \Delta v\right) = s_0 + max\left[0, Tv + \frac{v\Delta v}{2\sqrt{a_{max}b}}\right] \quad (9)$$

where, s_0 is the minimum desired distance (jam distance). T is the desired time headway (the time a driver wants to be behind the leading vehicle). b are the comfortable braking deceleration.

By introducing an internal dynamical variable to represent the subjective level of service, Treiber and Helbing [84] build the IDMM (intelligent driver model with memory) to incorporate memory effects in microscopic traffic models. The Multi-anticipative IDM models [85] consider essential finite reaction times, estimation errors, and spatial and temporal anticipation. Kesting et al. [86] enhanced the IDM model to assess the impact of adaptive cruise control vehicles on traffic capacity. Liebner et al. [87] applied IDM for ADAS to incorporate the spatially varying velocity profile to represent both car-following and turning behavior. Derbel et al. [88] modified the IDM to improve driver safety and respect vehicle capability in emergent situations by adding a new term cv^2/b to the expected safe time distance formula. Li et al. [89] modified IDM with power cooperation to strengthen the power of each vehicle in proportion to the immediately preceding vehicle. To build a realistic agent model, Eggert et al. [90] proposed the Foresighted Driver Model (FDM), which assumes that a driver acts in a way that balances predictive risk with utility. Milanés [91] calibrated CACC model and tested IDM on actual vehicles as reference for modeling ACC car-following behavior in traffic flow simulations. Hoermann et al. [92] extended the IDM for autonomous vehicles to describe a probabilistic motion prediction applicable to long-term trajectory planning. Treiber and Kesting [93] added external noise and action points to the IDM to study mechanisms behind traffic flow instabilities and indifferent regions of finite human perception thresholds. Zhou, Qu and Jin (2017) [94] added multiplication factors to build Cooperative Intelligent Driver Model(CIDM) to simulate AVs behavior to show that AVs could relieve traffic oscillation by give larger gaps for on-ramp vehicles. Hubmann et al. [95] extend the IDM model for complex urban scenarios by incorporating a behavior prediction framework with context-dependent upper and lower bounds on acceleration. A recent study [96] analyzed the limitations of the IDM model from the numerical analysis: the velocities of specific vehicles might become negative at specific times and might diverge to $-\infty$ in a finite time. To capture the stochastic nature of human driver behavior, IDM with time-varying parameters is proposed, with each parameter as a stochastic process and calibrating them in a time-varying manner [97]. The Self-Adaptive IDM (SA-IDM) [98] dynamically adjusts acceleration in response to the leading vehicle's speed and time headway, ensuring stable and overdamped platoon behavior under all conditions.

3) Psycho-physical Models:

Psycho-physical models are developed to approximate drivers' reasoning and decision process, considering the uncertainty and boundary effects in recognizing relative speeds and distances. Forbes [99] first introduced the human factor in car-following behavior and found that time headway should always be equal to or greater than perception-reaction time. Table III summarize two representative models: Action Point Model and Fuzzy Logic Model.

Action Point Model The Action Point Model (APM) is a variant of Signal Detection Theory (SDT) specifically applied to the context of car-following models in traffic psychology. The APM utilizes the principles of SDT to model the driver's perceptual sensitivity (signal detection) and response bias (decision criterion) in determining when to initiate an acceleration/braking action. By incorporating these components, the model provides a framework to understand and predict car-following behavior in terms of the driver's perceptual and decision-making processes. The prototype of APM research was presented by [100], where the first perceiving threshold that distinguished the boundary of whether the driver could perceive the velocity changing relative to the vehicle ahead was proposed. This threshold was given as the visual angle to the rear width of the vehicle ahead. The visual angel (θ) and its rate of change or angular velocity are calculated with the equations:

$$\theta = 2\tan^{-1}(\frac{w}{2H}) \tag{10}$$

$$\frac{d\theta}{dt} = -w * \frac{Vr}{H^2} \tag{11}$$

where H is the gap between leading and following vehicles. Vr is the relative speed. w is the width of the leading vehicle.

Todosiev [101] adopted a fundamental psycho-physical approach and investigated the thresholds where drivers change their behavior at an action point. Evans and Rothery [102] conducted experiments to quantify the thresholds between different phases. Wiedemann and Reiter [103] developed the well-known psycho-physical car-following model used in VISSIM. Another simulation program, PARAMICS, has incorporated the Fritzsche car-following model [104], an acceleration-based model from the psycho-physical car-following framework. As described in literature [105], once the absolute value of angular velocity exceeds its threshold, a driver notices that ego-vehicle speed is different from that of the leading vehicle and reacts with an ac/deceleration opposite in sign to that of the rate of change of visual angle. Winsum [106] integrated preferred time headway and Time-to-Collision (TTC) in a mathematical car-following model based on psychological evidence that human drivers regulate available time as a control mechanism. Wang et al. [107] devised ac/deceleration model based on the driver's cognitive mechanism using the Just-Noticeable Distance (JND) concept. Lochrane et al. [108] developed a multidimensional framework for modeling thresholds to account for the different behavior in the work and nonwork zones. Durrani and Lee [109] calibrated the eight Calibration Constants (CCs) built into the Wiedemann model used by VISSIM to quantify the threshold values for drivers' APs. Wagner et al. [110] analyzed a large dataset of car-following pairs to investigate the distribution of APs in terms of speed differences and gaps with SUMO's implementation of the Wiedemann model, finding that small distances indicate a slightly higher probability of finding an AP.

Fuzzy Logic Model To model human decision-making process, fuzzy logic allows the modeling of the estimates of a driver with vagueness, uncertainty, and subjectivity using fuzzy rules. Several studies [111]–[113] integrated the fuzzy theory into the GM model and attempted to "fuzzify" the

TABLE III
SUMMARY OF REPRESENTATIVE THEORY BASED MODELS FROM PSYCHO-PHYSICAL PERSPECTIVE

Model	Authors & Year	Key Strengths	Key Limitations
Action Points (AP)	Michaels and Cozen (1963) [100] Evans and Rothery (1973) [102] Wiedemann and Reiter (1992) [103] Fritzsche and Ag (1994) [104] Winsum (1999) [105] Wang et al. (2004) [107]	1. Human Perception-Reaction: Models can be more realistic compared to models based purely on physical or mathematical principles. 2. Adaptability: Action Point model can be adapted to represent different driver behaviors, which is particularly effective in simulating a variety of traffic components.	1. Complexity: Due to modeling multiple action points, it can be challenging for calibration. 2. Discontinuity: Discrete nature of action points can lead to abrupt changes.
Fuzzy Logic Models	Kikuchi and Chkkroborty (1992) [111] Chakroborty and Kikuchi (1999) [112] Zheng and McDonald (2005) [113] Chai and Wong (2015) [114] Hao, et al. (2016) [115] Bennajeh, et al. (2018) [116]	1. Human-like reasoning: Fuzzy logic can incorporate multiple inputs and outputs, allowing for various factors affecting driver behavior 2. Flexibility: Fuzzy logic is good at dealing with uncertainty and imprecision in human driving behavior.	1. Subjectivity: Creating an effective set of fuzzy rules and membership functions can be subjective. 2. Calibration and Validation Challenges: These models require extensive calibration and validation to ensure that the fuzzy rules accurately represent real-world driving behavior.

variables of the traditional GM model. A fuzzy cellular Automata Model [114] is developed to simulate the driver's decision-making process and estimate the effect of driving behavior on traffic performance, which utilizes fuzzy interface systems (FISs) and membership functions to simulate the perception-decision process of individual drivers. Hao et al. [115] incorporated fuzzy logic into the five-layer structure: Perception–Anticipation–Inference–Strategy–Action to build the CF model. Bennajeh et al. [116] employed four strategic variables, including gap distances, the velocity of followers and leaders, and the action of followers, to build the fuzzified anticipation car-following model. Although the fuzzy-logic model is a reasonable solution to consider the psycho-physical process, the sensitivity of membership values greatly impacts the accuracy and constructs of parameters relying on engineering preferences.

B. Adaptive Cruise Control Model

Unlike the previous models designed to simulate human driver behavior and reveal the disturbances of natural traffic, ACC models are developed to dampen the instability from the mechanistic view. ACC systems directly manage the basic operations of the vehicle, such as throttle control, braking, and steering. Low-level controllers are responsible for executing the commands to meet the higher-level objectives, like maintaining a set speed or following distance in ACC. Adaptive cruise control has been researched since the 1970s [117]–[122] and was first on the market when Mitsubishi launched the "Preview Distance Control" driver assistance system in 1995. ACC and CACC (Cooperative Adaptive Cruise Control) are heavily researched CAV applications. ACC models are designed for autonomous car following to reduce the feasible time headway among vehicles [123]. This paper [124] examines challenges of CACC systems in real-world deployment, including routing protocols, driver engagement, and adapting control designs to realistic traffic conditions. Some studies [125]–[127] suggest that the Constant Time Gap policy is more robust against error propagation through traffic than the Constant Distance Gap policy. By taking into account underlying the vehicle dynamics, e.g., force, torque, and power, Rakha et al. [128], [129] combined vehicle powertrain model with driver behavior to realize more realistic modeling of intelligent vehicle applications. Fadhloun-Rakha (FR) model [130], [131] was proposed by capturing heterogeneous driving behavior and vehicle engine dynamics, considering two constraints regarding car following interactions (collision avoidance and steady-state traffic stream). Table IV summarizes representative adaptive cruise control models.

1) Linear Model: Linear controller assumes acceleration is proportional to the deviation from target spacing and relative speed. Helly [132] proposed the first linear model that prescribes the adaptation of the acceleration of each vehicle as a function of relative velocity, relative distance from the leading car, and other pertinent parameters. Milanés et al. [133] compared four control techniques: PI, intelligent PI, fuzzy controller, and adaptive-network-based fuzzy control, showing that the intelligent PI performs best. Dias et al. [135] used a linear longitudinal force model as a feedforward controller or Proportional-Integral-Derivative (PID) to compensate for the powertrain non-linearity. The LQ-based controller has been applied on ACC and CACC systems and tested under rural and urban roads in a CV environment, considering both communication delay and driver reaction time [134], [136]. A classical Constant-Time Headway Relative-Velocity (CTH-RV) model [137] with gains of speed error and spacing error is described as follows.

$$a_i\left(t+\tau\right)=k_1\left(v_{i-1}\left(t\right)-v_i\left(t\right)\right)+k_2\left(r\left(t\right)-r_d(t)\right) \quad (12)$$

$$r_d(t)=t_g*v_i(t) \quad (13)$$

$$a_i\left(t\right)\in\left[d_{max},\ a_{max}\right] \quad (14)$$

where $k_1 > 0$ and $k_2 > 0$ are the gains of speed error and spacing error; v_{i-1} and v_i is the speed of the Preceding Vehicle (PV) and Following Vehicle (FV); $r(t)$ is the current distance between vehicles; $r_d(t)$ is the desired distance between vehicles; t_g is the desired time gap. a_{max} is the

maximum allowed acceleration; d_{max} is the maximum allowed deceleration. The acceleration and deceleration values are bounded by comfortable values at $2m/s^2$ and $-3m/s^2$.

By considering the V2V "hop-by-hop" communication, an effective CACC controller [138] received information about the velocity of PV, acceleration of PV, and braking capability of PV, which is defined as follows:

$$a_i(t+\tau) = k_0 \times a_{i-1}(t) + k_1 \times (v_{i-1}(t) - v_i(t)) + k_2 \times (r(t) - r_d(t)) \tag{15}$$

$$a_i(t) \in [d_i, a_{max}] \tag{16}$$

where d_i is the braking capability of FV. The desired spacing r_d is defined by the maximum safe following distance $r_{safe}(t)$, following distance with $0.5s$ time gap $r_{0.5s}(t)$, and a minimum allowed distance r_{min} that is chosen to be 2m.

$$r_d(t) = max(r_{safe}(t), r_{0.5s}(t), r_{min}) \tag{17}$$

$$r_{safe}(t) = \frac{v_{i-1}^2(t)}{2}(\frac{1}{b_i} - \frac{1}{b_{i-1}}) + \delta \times v_{i-1}^2(t) \tag{18}$$

$$r_{0.5s}(t) = 0.5 * v_i(t) \tag{19}$$

$$s_{min} = 2 \tag{20}$$

where, r_d, r_{safe}, and $r_{[t]s}$ are desired, safety, and spacing with t seconds gap, $\delta = 20$ ms is communication delay, b_{i-1} is the braking capability of the PV, b_i is the braking capability of the FV. The time gap ranges from $[0.5s, 1.4s]$. $1.4s$ is used when the preceding vehicle is not CACC-controlled.

2) Nonlinear Model: A Fuzzy ACC controller [139], whose output variables can be applied to low level controllers directly, was developed to ensure that the vehicle not only responds effectively to dynamic driving conditions but also respects and adapts to the driver's preferences and inputs, maintaining a balance between automation and human control. Naranjo et al. [140] developed an adaptive fuzzy controller for speed and distance vehicle control, comprised of three layers: Mechanical Layer, Electronic Layer and Control Layer. Pérez et al. [141] developed a CACC model through fuzzy controllers to perform platooning in highway scenarios. Zhang and Orosz [142] developed a distributed nonlinear controller for cooperation in multi-agent networks considering consensus and disturbance attenuation. To address the difficulties of obtaining parameters of the detailed longitudinal car-following model and account for nonlinear effects, Azızıaghdam and Alankuş [143] designed a two-loop controller that only uses parameters known to OEMs (Original Equipment Manufacturer). The outputs of the Inner loop are the accelerator pedal position sent to the vehicle CAN-Bus. The outer loop controls the relative distance and the relative velocity. A personalized adaptive cruise control (PACC) systems [144] use a Gaussian Process (GP) model to learn individual driving styles by implicitly considering the delay of driver reaction and the low-level vehicle dynamics. It significantly reduces human overrides in simulations, enhancing driving comfort and user experience compared to standard ACC models.

3) Model Predictive Control (MPC): This type of control algorithm is flexible and usually defined by multiple objective functions or cost functions that can directly incorporate key constraints. According to the spacing-control laws, Bageshwar et al. [145] applied the MPC in a vehicle cut-in scenario for transitional maneuvers of ACC vehicles. Based on radar measurements and V2X communication, Moser et al. [146] developed a stochastic MPC with a linear cost function using a piecewise linear approximation of the fuel consumption map. Cheng et al. [150] developed a multiple-objective MPC model integrated with direct yaw moment control (DYC) to ensure vehicle dynamic stability and improve driving comfort. A generic control framework proposed by [147], [148] assumes that accelerations of cruising vehicles are controlled to optimize a cost function reflecting different control objectives, i.e., safety, comfort, efficiency, and sustainability. Given that the control input $\mathbf{u} = \{u_i(\tau) | \tau \geq t_k\}$ is applied, the cost function of car-following is formulated as:

$$J(t_k, \mathbf{x}(t_k)|\mathbf{u}) = \int_{t_k}^{t_k+T_p} e^{-\eta\tau}\mathcal{L}(\mathbf{x}(t), \mathbf{u}(t), \tau)d\tau + e^{-\eta(t_k+T_p)}\phi(\mathbf{x}(t_k+T_p)) \tag{21}$$

The notation $J(t_k, \mathbf{x}|\mathbf{u})$ represents the expected cost associated with the control input $\mathbf{u}$ when the system states evolve from $\mathbf{x}(t_k)$ to $\mathbf{x}(t_k+T_p)$. Here, T_p denotes the prediction horizon. The function $\mathcal{L}(\cdot)$ represents the running cost during the interval $[\tau, \tau+d\tau)$, while $\phi(\cdot)$ represents the terminal cost, capturing the remaining cost at the terminal time. The parameter $\eta \in [0, +\infty)$ serves as a discounting factor with units of s^{-1}, balancing the current and future costs. Rolling horizon stochastic optimal control strategy for both ACC and CACC were implemented in a model predictive control fashion, considering the Uncertainties in system dynamics and sensor measurements [149]. A serial distributed model predictive control (MPC) [151] is developed for connected automated vehicles (CAVs), ensuring local stability and multi-criteria string stability by formulating future state constraints and tuning weight matrices, with mathematical proofs and simulations demonstrating its superiority over traditional MPC methods in maintaining stability.

III. Learning-based Models

Instead of an analytic modeling framework, more recently, data-driven car-following models have achieved human-level or beyond human-level performances under complex driving environments. It is unattainable to devise a parametric model considering all heterogeneous traffic compositions, road conditions, and varying data inputs. Whereas, data-driven approaches excel in scenarios with heterogeneous traffic compositions, diverse road conditions, and varied data inputs, not only in standard lane-based traffic situations but also in more challenging lane-free environments and in interactions with pedestrians. Table V summarizes representative data-driven models.

A. Supervised Learning Models

1) Machine Learning Model: With the availability of high-resolution vehicle trajectory data and the rampant develop-

TABLE IV
ADAPTIVE CRUISE CONTROL CAR-FOLLOWING MODELS

Model	Authors & Year	Key Strengths	Key Limitations
Linear Model	Helly (1959) [132] Milanes et al. (2012) [133] Kim et al. (2012) [134] Dias et al. (2014) [135] Jin and Orosz (2016) [136]	1. Flexibility: The linear control model can be designed to represent various traffic conditions by adjusting the gain values for different control inputs. 2. Robustness: The combination of feedback and feed-forward control mechanisms improves the model's robustness to uncertainties and disturbances in the environment.	1. Limitation of linearity: The model assumes a linear relationship between input variables (e.g., relative distance, speed, acceleration) and control actions (e.g., desired acceleration). With a time-invariant feedback gain, the control action does not adapt to changing conditions in the system's environment. 2. Safety Risks: Linear control systems often do not explicitly incorporate collision-free constraints, which are critical in ensuring safety in traffic management and automated driving systems.
Nonlinear Model	Holve, et al. (1995) [139] Naranjo, et al. (2003) [140] Perez, et al. (2013) [141] Zhang and Orosz (2016) [142] Aziziaghdam and Alankus (2022) [143] Wang, et al. (2022) [144]	1. Robustness: Nonlinear control techniques offer increased robustness to uncertainties and disturbances in the environment, such as sensor noise, communication delays, or changing road conditions. 2. Stability and performance guarantees: Some nonlinear control techniques provide stability and performance guarantees for the closed-loop system, which are crucial in safety-critical applications.	1. Design Complexity: Nonlinear models can be more complex, requiring extensive calibration, validation, and tuning efforts. The nonlinear relationship between inputs and outputs makes it harder to understand the underlying driving behavior or control actions.
Model Predictive Control	Bageshewar, et al. (2004) [145] Hoogendoorn, et al. (2012) [147] Wang, et al. (2013) [148] Moser, et al. (2015) [146] Zhou et al. (2017) [149] Cheng, et al. (2019) [150]	1. Optimal control: MPC provides an optimal control solution by minimizing a cost function that can incorporate various objectives, such as maintaining a safe distance, reducing fuel consumption, or improving driving comfort. 2. Explicit constraints handling: MPC can handle constraints explicitly, such as acceleration limits, speed limits, or safe following distances. 3. Adaptability: MPC-based models can be adapted to different driver behaviors or customized driving styles by adjusting the cost function or model parameters.	1. Computational demands: The optimization problem in MPC needs to be solved in real-time, which can be computationally intensive, especially for large prediction horizons or high-dimensional models. 2. Prediction model accuracy: The performance of MPC models highly depends on the accuracy of the prediction model used to predict the future states of the leading vehicle and the following vehicle.

ment of machine learning (ML) algorithms and tools, several high-accuracy machine learning and deep learning algorithms have been developed. Kehtarnavaz et al. [152] developed a time-delay neural network (TDNN) autonomous vehicle following module to replace a conventional proportional and derivative/proportional and integral (PD/PI) controller and implemented it on actual vehicles. Panwai and Dia [153] developed a neural agent car-following model for mapping perceptions to actions. They compared the speed and position of individual vehicles from data to model output trajectory profiles. Khodayari et al. [154] developed an Artificial Neural Network (ANN) to simulate and predict the car-following behavior based on the driver-vehicle unit's reaction delay, and input parameters include relative speed, relative distance, and follower's speed. Zheng et al. [155] developed ANN to output the following vehicle's speed by considering the driver–vehicle reaction delay in relative speed and acceleration, the gap, and the vehicle's speed. Wei and Liu [156] proposed a Support Vector Machine approach to investigate the asymmetric characteristics of car-following behavior and its impact on traffic flow evolution. The Support Vector Machine model takes space headway, velocity, and relative speed as the inputs and the output is the follower's speed. Another study [157] proposed a data-driven locally weighted regression approach for optimizing car-following model estimation by applying an existing technique and calibrated using instrumented vehicles in Naples, Italy. The gradient boosting of regression tree (GBRT) algorithm [158] was applied to NGSIM data to develop a car-following model that outperforms traditional GHR models in capturing the motion characteristics of successive vehicles, using cross-validation and sensitivity analysis for parameter tuning. Yang et al. [159] developed a hybrid method by integrating theoretical-based and machine-learning models to obtain better accuracy while maintaining the physical meaning. Kamjoo et al. A Gaussian Process Regression Approach [160] to model individual longitudinal driving behaviors by integrating parametric and non-parametric methods, enhancing predictions with large data sets and adapting to new variables without changing the model structure. [161] implemented Support Vector Machine and Long-Short-Term-Memory (LSTM) models to investigate the winter operation's impact on car-following behavior, showing that the presence of snowplows leads to significantly different car-following behaviors.

2) Deep Learning Model: Driving behaviors can be treated as sequential data since the driver's following action is conditioned on their previous action. Deep Learning (DL) models, particularly Recurrent Neural Networks (RNN), are similar to humans in memory-based decision-making. Morton et al. [162] evaluated LSTM networks for simulating driver acceleration behavior on highways, which can effectively replicate real driver behavior in traffic models, matching or surpassing baseline methods in accuracy and realism. Zhou et al. [163] developed an RNN with inputs of the gap, relative speed, and follower's speed to predict acceleration, speed,

TABLE V
SUMMARY OF REPRESENTATIVE DATA-DRIVEN CAR FOLLOWING MODELS

Model	Authors & Year	Key Strengths	Key Limitations
Machine Learning	Kehtarnavaz et al. (1998) [152] Panwai and Dia (2007) [153] Khodayari et al. (2012) [154] Zheng et al. (2013) [155] Wei and Liu. (2013) [156] Papathanasopoulou and Antoniou (2015) [157] Dabiri and Abbas (2018) [158] Yang et al. (2018) [159] Soldevila et al. (2021) [160] Kamjoo et al. (2022) [161]	1. Interpretability: Traditional ML models (e.g. Linear Regression, Decision Tree and SVM) are interpretable, making it easier to understand how decisions are being made, which is crucial for safety-critical applications. 2. Faster Calibration and Inference: Due to their simpler nature, traditional ML models can be trained faster and often have lower latency during inference.	1. Feature Engineering: ML models often rely heavily on manual feature engineering. 2. Scalability: As the complexity of the scenario increases, the performance of traditional ML models might not scale as well as other models.
Deep Learning	Morton et al. (2016) [162] Zhou et al. (2017) [163] Wang et al. (2017) [164] Wang et al. (2019) [165] Wu et al. (2019) [166] Zhang et al. (2019) [167] Ma and Qu (2020) [168] Zhou et al. (2022) [169] Xu et al. (2023) [170] Lu et al. (2023) [171] Xu et al. (2024) [172] Chen et al. (2024) [173]	1. Data-driven: DL models learn directly from data, allowing them to capture the complexity of real-world driving behavior without the need for explicit modeling assumptions. 2. Flexibility: A wide range of DL techniques can be used, enabling the selection of an appropriate model for the specific problem and data. 3. Incorporation of driver characteristics: DL models can learn to capture driver characteristics or latent driving styles, enabling the modeling of heterogeneous human driving behaviors.	1. Labeled data requirements: Supervised learning models require a large amount of labeled training data. 2. Cascading errors: In sequential prediction tasks, an incorrect estimation at an early stage can propagate through subsequent steps, leading to increasingly inaccurate decisions.
Imitation Learning	Shimosaka et al. (2014) [175] Shimosaka et al. (2015) [176] Kuefler et al. (2017) [177] Gao et al. (2018) [178] Zhou et al. (2020) [179] Zhao et al. (2022) [180]	1. Capturing implicit knowledge: Imitation can capture the implicit knowledge and preferences of human drivers, such as safety, comfort, and efficiency, without the need to explicitly specify these objectives. 2. Robustness: By learning from expert demonstrations, Imitation models can be more robust to environmental uncertainties and disturbances, as the expert's behavior may already account for these factors.	1. Quality of expert demonstrations: The effectiveness of IL depends on the quality and diversity of the expert demonstrations provided. 2. Ambiguity in Learned Policy: It is important to note that if multiple behaviors could result in the same observed behavior, IL might not distinguish the true intent behind the actions.
Unsupervised Learning	Koutsopoulos and Farah (2012) [181] He, et al. (2015) [182] Lefevre et al. (2015) [183] Guha et al. (2022) [184]	1. Data Efficiency: Unsupervised learning (UL) can utilize large amounts of unlabeled data, which are more abundant and less costly to obtain than labeled data. 2. Anomaly Detection: UL is particularly good at detecting outliers or anomalies, which could be crucial for identifying unexpected behaviors or conditions. 3. Clustering Behaviors: UL can cluster similar driving behaviors, which could be used to understand different driving styles and adapt the car-following behavior accordingly.	1. Validation Challenges: Without labeled data, it can be challenging to validate the model's predictions or to determine its accuracy in a traditional sense. 2. Risk of Overfitting: There's a risk that the model will overfit to the noise in the data, detecting spurious patterns that do not generalize well to unseen situations.
Reinforcement Learning	Ioannou and Chien (1993) [185] Desjardins and Chaib-draa (2011) [186] M. Zhu et al. (2018) [187] Y. Zhang et al. (2019) [188] M. Li et al. (2020) [189] Yen et al. (2020) [190] Masmoudi et al. (2021) [191] Yavas et al. (2022) [192] Wang et al. (2022) [193] Tang et al. (2022) [194] Liao et al. (2024) [195]	1. Adaptability: RL is highly adaptable to a range of conditions and can improve its policy based on the feedback from the environment. 2. Optimization: RL agents can learn to optimize multiple objectives simultaneously, balancing trade-offs between safety, efficiency, and comfort. 3. Exploration vs. Exploitation: RL allows the algorithm to discover new strategies while still leveraging what has been learned.	1. Training stability: RL algorithms can be sensitive to hyperparameters and prone to instabilities during training, requiring careful tuning and design choices. 2. Computational complexity: RL often requires significant computational resources for both training and inference, which can be a limiting factor. 3. Safety concerns: RL agents may explore suboptimal or unsafe actions that could lead to accidents or undesirable behaviors during the learning process as a concern in safety-critical applications.

and gap that can be used for car-following control. Wang et al. (2017) [164] developed deep learning-based car-following with Gated Recurrent Unit (GRU) that embeds model outputs' prediction and memory effects. Wang et al. (2019) [165] applied Gated Recurrent Unit (GRU) with different timescale historical information as inputs to test memory effects on car following behavior. Wu et al. (2019) [166] built a deep learning-based model to mimic human drivers' memory, attention, and prediction (MAP) mechanisms. Zhang et al. [167] built an LSTM model with a hybrid retraining constrained (HRC) training method to model car-following and lane-changing behaviors. Given that CF behaviors (acceleration, deceleration, and cruising) are made for continuous time steps rather than step-by-step with memory effects and delayed

reaction time, Ma and Qu [168] developed a seq2seq LSTM model for predicting multistep car-following behaviors. Their model inputs include gap distance, relative speed difference, and the speed of the subject vehicle, and the model output is the multiple-step accelerations/decelerations. Zhou et al. [169] applied a transfer learning-based LSTM car-following model for adaptive cruise control to address the ACC data scarcity problem by transferring useful features from human-driven data. Missing data in vehicle state information often leads to gaps in the data required for making accurate control decisions. A longitudinal CF model leverages Transformer-Generative Adversarial Networks (TransGAN) [170] for this purpose, which is particularly significant in scenarios where data acquisition devices malfunction. An improved sequence-to-sequence model for connected automated vehicles (CAVs) [171] enhances the accuracy of simulating speeds and positions by incorporating kinematics information from multiple preceding vehicles and using a novel deep learning framework. Another sequence-to-sequence model [172] incorporates an attention mechanism to learn the probability distribution of driver reaction delay. Aggressiveness Informed Car-Following (AICF) [173] was proposed approach, which integrates driving style as a dynamic input in data-driven car-following models to address the limitations of using predefined patterns and categories.

3) Imitation Learning Models: Imitation learning (IL) a subset of machine learning focused on learning from demonstration, which has several advantages: 1. It can meet the preferences of driving styles; 2. Other drivers would easily understand the automated car-following maneuver. Behavior Cloning (BC) models, which are the simplest forms of Imitation Learning, try to learn the maneuver of the human driver given the expert's demonstrations that are divided into state-action pairs. The BC model can discover complex car-following strategies for nonlinear and high-dimensional systems. Inverse reinforcement learning (IRL) can be seen as a form of imitation learning where the model is not just learning to mimic the actions of the expert, but is actually trying to understand the motivations (i.e., the reward function) behind those actions. This can lead to more robust and versatile models, as they are not just copying behavior, but are learning the principles that guide that behavior. With naturalistic driving behavior data, the IRL would approximate that human's reward function for the driving task. Instead of learning a policy from a reward function, Inverse Reinforcement Learning (IRL) is trying to learn a reward function from the demonstration of a task [174]. Shimosaka et al. [175] proposed an IRL framework to model risk anticipation and defensive driving, which has a better long-term predict [176] developed multiple reward functions with clustering of environments to better handle environmental diversity. The framework then simultaneously learns the parameters specific to each cluster. This method significantly improves the model's adaptability and accuracy in diverse settings. Kuefler et al. [177] proposed Generative Adversarial Imitation Learning (GAIL) framework to imitate human car-following behavior with three non-linearity in mapping from states to actions, high-dimensional state representation, and stochasticity. Gao et al. [178] proposed a car-following algorithm with a reward function based on IRL under different conditions. The experimental verification is conducted on a dynamic driving simulation test bench. Zhou [179] developed a Maximum Entropy Deep IRL approach using a neural network to estimate the rewards. Model input features include time headway, relative speed, and maximum speed, and model output is speed control. Zhao et al. [180] developed a Personalized-ACC framework using IRL, which classified the driver and weather conditions for the pre-trained off-line IRL model and compared it with the IDM model regarding safety and comfort measured by takeover percentage.

B. Unsupervised Learning Models

Unlike traditional approaches that rely on supervised learning with labeled datasets, unsupervised learning based car following models are designed to autonomously discover patterns and relationships within unlabeled traffic data. This paradigm shift facilitates a more organic understanding of traffic dynamics. As vehicular technologies and traffic systems evolve towards greater autonomy and complexity, unsupervised learning provides a crucial framework for developing more adaptive, efficient, and safer car following algorithms. A latent class model [181] is developed to estimate drivers' decision, based on the assumption that these decisions are probabilistic and the result of a number of explanatory variables. The latent class model is formulated as a discrete choice problem with systematic utility functions. A simplified k-nearest mean model [182] was developed that only takes the position as input and reproduces traffic dynamics without premises and calibration. This approach selects the most similar historical records and outputs the average of the k nearest neighbors. A learning-based architecture for autonomous driving which combines a driver model and a Model Predictive Controller (MPC) [183]. The driver model is built on Hidden Markov models (HMM) representation of human control strategies to imitate different driving styles. The MPC guarantees safety by enforcing a set of constraints on the vehicle state. A robust unsupervised learning method [184] is proposed to model temporal dynamic interaction between two agents comprises a pair of well-aligned trajectories, which provides more robust and data-driven learning approaches for highly diverse and dynamic interactions. The unsupervised learning framework doesn't rely on strong modeling assumptions that are often used to represent latent variables that govern the actions among agents and the driving environment.

C. Reinforcement Learning

In the development of advanced car-following models, Reinforcement Learning (RL) has emerged as an important approach due to its capacity for decision-making in complex and dynamic environments. RL can be broadly classified based on the knowledge of the environment: model-based RL, where the model of the environment is known, and model-free RL, where it is not. Furthermore, RL methods can also be categorized based on their use of the optimality condition to derive an optimal policy. Indirect RL methods leverage solutions from the Bellman or Hamilton-Jacobi-Bellman (HJB)

equations, founded on Bellman's principle of optimality, to ascertain at least one optimal policy. In contrast, direct RL methods avoid the optimality condition, opting instead to search the entire policy space to optimize the primal problem directly. By utilizing these RL strategies, car-following models aim to replicate and optimize human-like driving behaviors, focusing on safety, efficiency, and comfort while navigating the uncertainties of real-world traffic conditions. The RL agent chooses an action a from action space A based on the observed state s to get a reward $R(s,a)$. During the learning process, the agent uses an exploration-exploitation trade-off. It explores the environment by taking random actions to discover new states and learn more about their rewards. The agent gradually learns the optimal action policy for each state by repeatedly updating the Q-values based on observed rewards.

$$\begin{aligned}\text{New } Q(s,a) \leftarrow\ & Q(s,a) \\ & + \alpha\left[R(s,a) + \gamma \max Q'(s',a') - Q(s,a)\right]\end{aligned} \tag{22}$$

where α is the learning rate, and γ is the discount rate. $Q'(s',a')$ is next Q-value for all possible s' and a'. $R(s,a)$ is the reward function.

Ioannou and Chien [185] applied RL to develop an autonomous intelligent adaptive cruise control system. Desjardins and Chaib-draa [186] introduced a policy gradient algorithm estimation and used BPNN for longitudinal vehicle control. Their RL-based car-following control was implemented in the simulator to realize the efficient behavior of CACC. Zhu et al. [187] proposed a framework for developing human-like car-following based on deep reinforcement learning using speed deviations as a reward function. Zhang et al. [188] developed a deterministic promotion RL algorithm (DPRL) based on an actor-critic frame and the policy gradient method for longitudinal velocity control, implemented in CarSim simulated environment and real-world experiment. Li et al. [189] proposed a deep deterministic policy gradient (DDPG)-based driving strategy taking information from in-vehicle sensors for system performance optimization. Yen et al. [190] developed a proactive car-following model using deep-RL considering road efficiency, safety, and comfort, which shows better performances in terms of time heady, time to collision (TTC), and jerk (rate of change in acceleration). Yavas et al. [192] developed a Model-Based Reinforcement Learning (MBRL) for ACC control with a hybrid policy that combines the Intelligent Driver Model following policy with the Deep Reinforcement Learning policy. Wang et al. [193] devised an RL model for longitudinal velocity control in a three-vehicle mode based on the safety level, consisting of the collision avoidance priority of the leading and following vehicles and the expected acceleration/deceleration decision. Most CF models take relative speed, acceleration, and gap as inputs, while some studies [191], [194] use video object detection to learn the driving strategy. Such models demonstrate promising results, which can efficiently be deployed on embedded devices for advanced driver assistance systems. This study [195] develops a personalized car-following model using a memory-based deep reinforcement learning approach, combining Twin Delayed Deep Deterministic Policy Gradients (TD3) with Long Short-Term Memory (LSTM) networks to optimize car-following behavior based on diverse driving styles for improved safety, efficiency, and comfort, demonstrating superior performance and convergence compared to other models.

IV. Knowledge-driven Models

Knowledge-driven methods leverage extensive pre-trained Large GenAI models and foundational knowledge to generate new content and solutions, simulating commonsense-based understanding and reasoning that reflect deep concepts. Large GenAI models can address the heterogeneity in traffic components, varying road environments, and the stochastic nature of driving with an agent-based approach. CF behavior is considered as a subtask of Autonomous Driving (AD), which suggests that large GenAI models developed in AD can also be applicable in CF applications. The kinematic CF models (e.g., IDM models) are useful as baseline for validating new knowledge-based driving agents. In CF scenarios, these knowledge-driven methods reason about vehicle behaviors, enhancing decision-making in complex driving scenarios. Additionally, they provide context-aware systems that understand and react to real-time driving conditions, leading to more realistic simulations, optimal control, and enhanced human-machine interaction.

A. Large Language Models (LLMs)

LLMs can generate detailed driving instructions and responses based on signal inputs and human-like understanding. These models can create complex driving policies by predicting and generating the next action based on previous driving scenarios and contextual data, which enhances the decision-making processes in autonomous CF, making vehicles more adaptive and responsive to dynamic driving environments. Examples of LLM-based autonomous driving models include DriveLLM [196], DiLu [198], Driving with LLMs [197], and Drive-as-you-speak [199].

B. Vision-Language Models (VLMs)

VLM models integrate visual inputs, such as images and videos, with textual descriptions to enhance understanding and generate new insights or content. These models excel in scenarios that require multi-modal perception and comprehension, enabling accurate interpretations of complex driving situations. Representative models include vision-language-action driving model (VLAM) (LINGO-1 [200] & LINGO-2 [201]) and DriveVLM [202].

C. Multimodal Large Language Models (MLLMs)

MLLMs extend the capabilities of traditional LLMs and VLMs by integrating multiple types of data inputs. These models can process and understand a combination of various sensor input to provide a holistic view of the driving environment. A recent survey paper [203] on Multimodal Large Language Models for AD sheds light on the latest advancements and research directions in the field. Representative models include DriveMLM [204] and LMDrive [205].

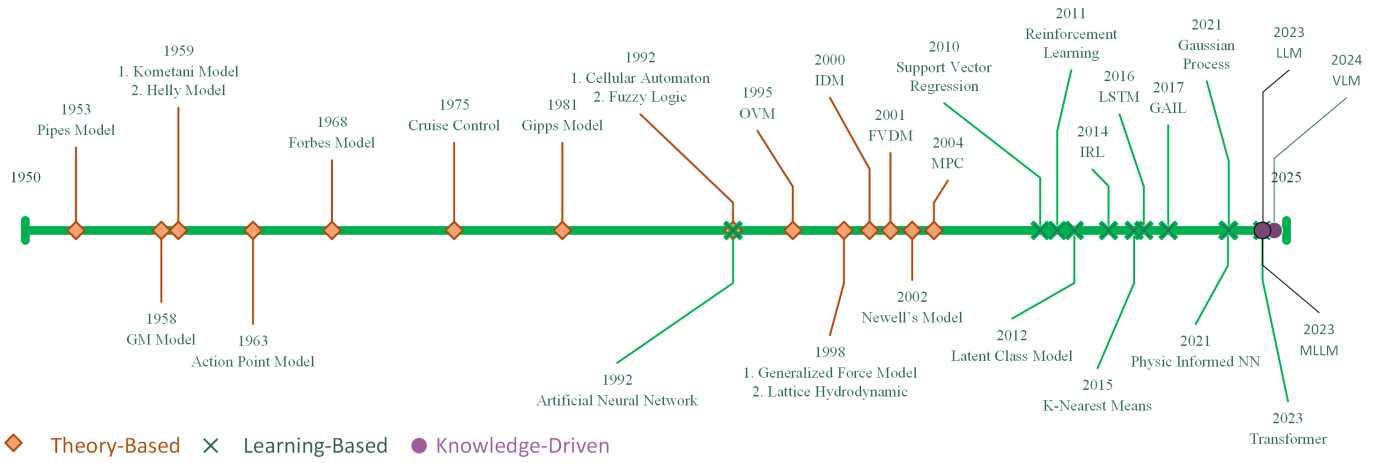

Fig. 3. The Historical Timeline of Reviewed Master Models and Key Variants.

V. Discussion

In this review paper, the focus on master models (Figure 3) offers a structured approach by providing a clear framework for categorization and understanding. The Figure 4 illustrates the efficacy of these master models in delineating the progression and branching out of various derivative algorithms within the field. By doing so, it highlights the master models' role as pivotal reference points from which the landscape of car-following behavior can be surveyed. Automatic literature tools [206] adeptly navigate the extensive body of literature to surface the most relevant studies, demonstrating the significant advantage of leveraging structured categorizations to manage and interpret the wealth of research in this evolving area of traffic dynamics and control systems. This paper's focused review through the lens of multi-disciplines thus not only consolidates existing knowledge but also guides future investigations.

A. Theory-Based vs. Learning-Based vs. Knowledge-Driven

Decades of theory-based driving behavior research are now merging with learning-based and knowledge-driven opportunities. Although the data-driven models will undoubtedly become critically important, the traditional driver behavior expertise remains relevant. It is auspicious to integrate rule-based models with the fast-evolving AI models, which can reason, learn, and fill in missing information to achieve better interpretability, generalizability, and reliability.

Several hybrid CF models were developed by integrating rule-based and AI-based models, which leverages the strengths of different approaches, enhancing accuracy, robustness, and safety in systems like autonomous vehicle control, where both predictable behavior and adaptability to real-world complexity are crucial. Li et al. [207] utilized the adaptive Kalman filter algorithm to integrate the LSTM data-driven model with the IDM theoretical-driven model to build the car-following model. Mo et al. [208] designed Physic-Informed Deep Learning architectures encoded with IDM and OVM to predict accelerations in four traffic regimes: acceleration, deceleration, cruising, and emergency braking. This paper [209] focused on analyzing the stochastics and heterogeneity in traffic dynamics using a data-driven car-following model. It examined the shape of the distribution of a stochastic term in the car-following model, predicting accelerations accurately and identifying relationships between predicted distributions' modality, standard deviation, and traffic states. This research [210] introduces the IDM-Bi-LSTM combination car-following model, which integrates the theory-driven Intelligent Driver Model (IDM) with a data-driven model optimized by Particle Swarm Optimization (PSO) to enhance trajectory prediction accuracy and maintain controllability and security. Numerical simulations demonstrate that this hybrid model significantly reduces prediction errors and outperforms both individual IDM and PSO-Bi-LSTM models in simulating heterogeneous and asymmetric driving behaviors.

The comparison emphasizes (Table VI) the contrast between Rule-based and AI-based approaches in modeling car-following behavior. While theory-based models offer a clear, predictable framework grounded in established principles, their limitations in handling the complexity and variability of real-world driving scenarios highlight the potential benefits of incorporating learning and commonsense knowledge. Conversely, AI-driven models' ability to adapt to and learn from diverse environments is tempered by challenges related to interpretability, the quality of learned behavior, and the handling of rare but critical situations. Balancing these strengths and limitations often results in more reliable, accurate, and practical systems for real-world applications.

B. Applications

Analyzing car-following behavior and developing intelligent control algorithms have significant consequences for road safety and energy efficiency. The profound impacts of the different driving algorithms (e.g., strategic models, model-predictive control, RL controller, and LLM-agent) and various connected environments have set the stage for a new era of traffic flow dynamics. Many applications based on car-following models have been undertaken, summarized with the following topics.

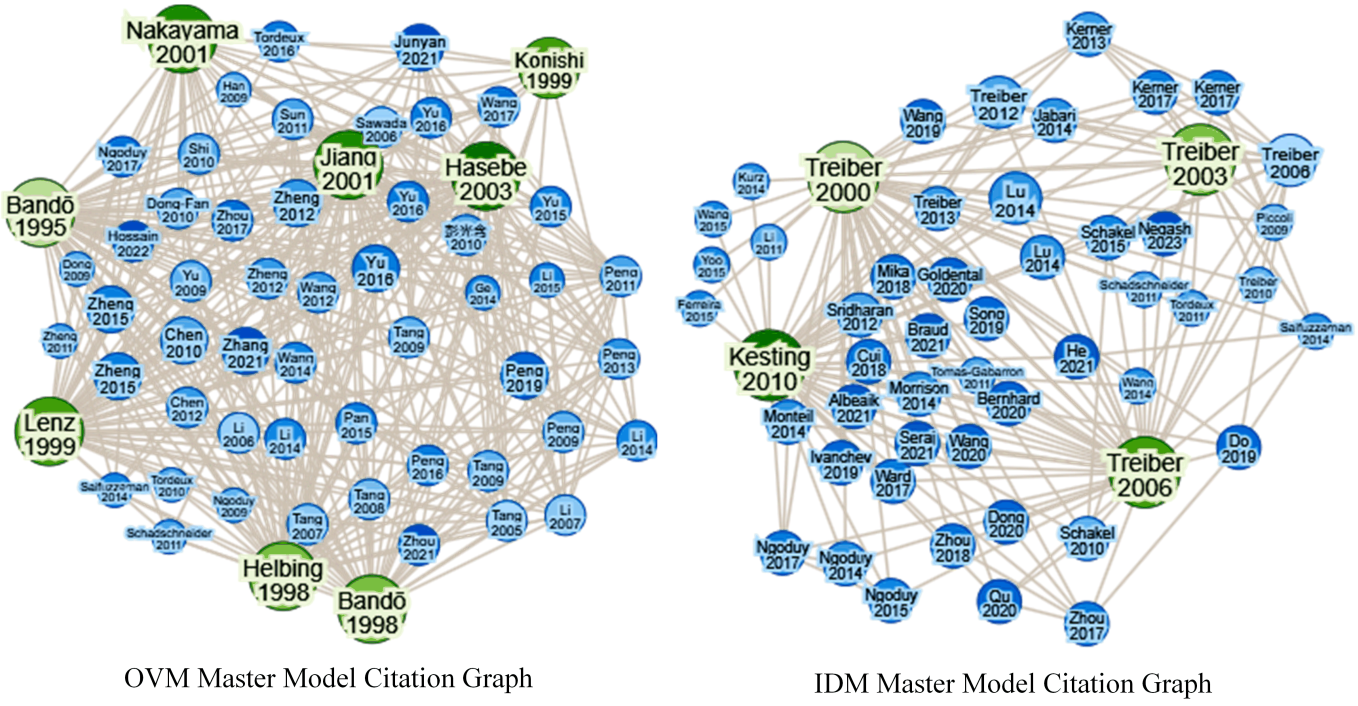

Fig. 4. Navigating the Literature with Reviewed Master Models Using OVM and IDM as Examples

TABLE VI
COMPARATIVE OVERVIEW OF THEORY-BASED MODELS VS LEARNING-BASED MODELS VS KNOWLEDGE-DRIVEN MODELS

Theory-Based Models	**Learning-Based Models**	**Knowledge-Driven Models**
Strengths	**Strengths**	**Strengths**
• Solid Theoretical Foundation: Built on well-established principles and theories, offering predictable behavior within their domain of validity. • Interpretability: Due to their reliance on explicit formulas and parameters, these models provide clear insights into the cause-and-effect relationships within the modeled phenomena. • Generalizability: Capable of generalizing across different scenarios, ensuring that the model adheres to expected physical laws and behavioral norms, which helps in maintaining performance even under unusual conditions not covered by the data.	• Accuracy and Prediction: Learning-Driven models are adept at capturing non-linear patterns from large datasets. This can lead to highly accurate predictions based on real-world data. • Adaptability to Complex Environments: Ability to achieve human-level or superior performances under diverse driving conditions. • Handling of Heterogeneous Data: Efficient processing and learning from varied data inputs, adapting to different traffic and infrastructure conditions.	• Physical-Social Intelligence: They understand both physical laws and drivers' expectations and emotional needs, enabling them to identify and comprehend various patterns, rules, and relationships in driving behavior. • Handling Edge Cases: Knowledge-Driven Large GenAI models address edge cases and unusual scenarios by leveraging commonsense reasoning and proactive decision-making. • Context-Aware Systems: They enable context-aware systems that understand and react to real-time driving conditions, enhancing decision-making, human-machine interaction, and the overall safety and efficiency of intelligent vehicles.
Limitations	**Limitations**	**Limitations**
• Complexity of Traffic Dynamics: Struggles to account for all heterogeneous traffic compositions, infrastructure conditions, and varying data inputs. • Parametric Restrictions: Limited by the need to pre-define parameters, which may not capture the full spectrum of driver behaviors or environmental conditions.	• Overfitting and Scalability: Due to environmental diversity and sensor variability, learning-based techniques perform well on the data on which they were trained but not always adaptable on new, unseen datasets or in more complex environments. • Learning Undesirable Behaviors: Potential to learn from bad behaviors due to drivers' unpredictability or poor data quality.	• Alignment Challenges: Knowledge-Driven pre-trained foundational models are end-to-end systems. Misalignment occurs when objectives and behavior do not align with the intended use case, leading to undesired consequences or vulnerabilities. • Safety and Ethical Concerns: Addressing potential biases, prioritize safety, and ensuring transparency in decision-making, are ongoing challenges that need careful consideration.

- **Topic 1:** Traffic Flow Dynamics. Simulation analysis through car following models offer insights into emerging traffic flow characteristics under conventional or connected environment. CF models serve as the basis for exploring how connected and automated mobility (CAM) could enhance traffic dynamics and address both the opportunities and challenges presented by mixed traffic scenarios [211]–[223].
- **Topic 2:** Road Safety and Energy Efficiency. Car-following models play a crucial role in developing eco-driving strategies and improving safety levels on the road. By simulating the behavior of vehicles in various traffic conditions, these models can help identify optimal driving patterns that minimize energy consumption and emissions while also reducing the risk of accidents. As research in this area continues to evolve, more sophisticated systems will be evaluated and developed with CF models to enhance both environmental sustainability and road safety. [224]–[229]
- **Topic 3:** V2X and Wireless Communication. Car-following models can significantly enhance wireless communication applications within the Internet of Vehicles (IoV) to improve situational awareness. CF models facilitate the development of algorithms that optimize data transmission and reception among connected vehicles, improving the efficiency of vehicular ad-hoc networks (VANETs). This fosters the deployment of advanced IoV services, and the delivery of various Internet-based applications to drivers and vulnerable road users (VRU), thereby leveraging the full potential of cloud computing and IoT technologies in the automotive domain [230]–[237].
- **Topic 4:** Innovative Traffic Control. Researchers have focused on traffic control and novel operation strategies. CF models are employed to optimize the control algorithms, such as collaborative driving strategy, traffic signal timing, shock-wave damping, or variable speed limits [238]–[247].
- **Topic 5:** Cyber-Physical Human Systems. The seamless integration of human elements, computer-based algorithms, and physical components is essential to transform transportation systems into Cyber-Physical Human Systems (CPHS). By leveraging car following models, researchers are able to develop analytical frameworks that not only enhance the performance of these vehicles but also improve the overall security of the transportation system. This integration and testing pave the way for the eventual realization of fully autonomous driving capabilities [248]–[256].
- **Topic 6:** Human Factors Research. Car following models are effective tools to incorporate psychological and behavioral aspects to assess human-machine interaction, driver behavior, and the potential consequences of automated systems. By understanding human elements and studying human-machine interfaces, researchers aim to improve the acceptance, trust, and performance of intelligent vehicles [257]–[263].
- **Topic 7:** Trajectory Reconstruction. Sensor-based trajectory acquisition often encounters issues with noise and missing information. This can be due to various factors such as sensor limitations (fixed location or penetration rates), environmental conditions, and the inherent complexity of traffic environments. This is where car following models become crucial to estimate the missing information in vehicle trajectories with the partial data collected from these sensors. The complete and continuous vehicle trajectory is vital for various applications. [264]–[275].

C. Dataset

Detailed high-quality vehicle trajectory data are of great significance for the development of driving behavior models and vehicle control algorithms. With realistic microscopic longitudinal and lateral maneuvers, the valuable vehicle trajectory data lay the foundation for many scientific discoveries. The collection of realistic vehicle trajectory data is identified as important as modeling and understanding the interactions of travelers. There are two types of vehicle driving datasets that are often used as benchmarks. 1. Lagrangian-type data collected from mobile devices on floating vehicles. 2. Eulerian-type traffic-flow data that monitors the entire traffic flow from stationary devices. An integrated Benchmark [276] for Car-Following Behavior Modeling is extracted from five public driving datasets under the same format and criteria, which consists of more than 80K car following events. A summary table of datasets suitable for car following model research is presented in Table VII. It is important to note that while this table showcases some of the most pivotal datasets, it is by no means exhaustive. There are undoubtedly other datasets available that could provide valuable contributions to the field. The datasets highlighted in our summary are primarily focused on open street environments capturing captures the complexity and unpredictability of naturalistic driving scenarios as opposed to closed testing environments.

VI. CONCLUSION AND OUTLOOK

This review aims to enhance understanding of the underlying design principles in different categories of CF Models, enabling researchers to formulate problems with a suitable framework. As the path to achieve level-5 autonomy continues, incorporating general traffic flow knowledge presents both challenges and opportunities. The future direction of driver behavior models, particularly in the context of integrating human-like reasoning capabilities and learning efficiencies into car following models, presents a fascinating and complex challenge. Bridging the gap between conventional fixed and myopic driver behavior models and rapidly advancing AI models requires a multifaceted approach.

Hybrid-learning Systems: The potential of hybrid-learning systems lies in their ability to produce models that are not only highly predictive but also deeply understandable and controllable. Among the leading methodologies in this frontier is Physics-informed Machine Learning (PIML), which stands out for its ability to harmonize traditional kinematic models with the implicit understanding gleaned from vast datasets of

TABLE VII
SUMMARY OF TRAJECTORY DATASET COLLECTED FROM BOTH MOBILE AND FIXED SENSORS

Dataset	Year	Highlight	Sample Size
Next Generation Simulation (NGSIM) [277]	2006	The NGSIM datasets provide an unprecedented depth of trajectory data, which were captured at four locations: The US-101 and I-80 sites offer insights into freeway conditions, while Peachtree Street and Lankershim Boulevard provide data on urban street environments.	Each dataset is segmented into three 15 minute periods, covering approximately 500 meters (1,600 feet) in length, with over 11.8 million rows of trajectory data.
Zen Traffic Data [278]	2018	Zen Dataset is characterized by its high traffic volume, two lanes in each direction, and a complex road layout featuring S-shaped curves and elevation changes.	3600 trajectories on 2km highway segment for 1 hour.
HighD [279]	2018	The HighD dataset uses drone on German highways over various sections of highways.	110,500 vehicles, 28,000 vehicle miles traveled, and 147 hours
ACFR five roundabouts dataset [280]	2019	This dataset is aimed at studying naturalistic driving behaviors at unsignalized intersections. The dataset offers insights into a wide range of driving behaviors, such as negligence, speeding, and assertiveness.	A dataset containing over 23,000 vehicle recordings at five different roundabouts, captured by a LiDAR-based detection and tracking system.
INTERA--CTION [281]	2019	The INTERACTION dataset stands out for complex and critical driving situations, often within unstructured settings lacking clear right-of-way rules, where driver behavior is significantly influenced by the actions of others.	Roundabout Scenarios (10,479 vehicle trajectories). Unsignalized intersection scenarios (14,867 trajectories). The merging and lane change scenarios (10,933 trajectories). The signalized intersection (3,775 trajectories).
inD [282]	2020	The inD (intersection Drone) dataset is a collection of naturalistic road user trajectories at urban intersections.	More than 11,500 road users, including vehicles, bicyclists and pedestrians at intersections with 10 hours of data from four intersections
rounD [283]	2020	The rounD dataset is a collection of road user trajectories at roundabouts in Germany, captured using a drone camera.	A total of 6 hours of recordings with more than 13,746 road users including cars, vans, trucks, buses, pedestrians, bicycles, and motorcycles.
I-24 Motion [284]	2020	I-24 Motion stands out for its large-scale data acquisition, capturing detailed vehicle trajectories across multiple lanes and over an extended period.	With 276 cameras that cover 4.2 miles of I-24, the data collected is intended to support a wide range of applications.
pNEUMA [285]	2020	Coordinated Drone video-based vehicle trajectory collected in downtown area in Athens, Greece during morning peak hour.	The pNEUMA dataset encompasses close to half a million vehicle trajectories over a congested 1.3 km² area, featuring more than 100 km-lanes or road network.
Ubiquitous Traffic Eyes [286]	2021	The database encompasses a wide array of road conditions and traffic conditions, with manual corrections ensured 100% vehicle detection and tracking accuracy.	The data segments vary in duration, ranging from 4 to 20 minutes. This dataset is characterized by its high precision, with time accuracy up to 0.1 seconds and position accuracy up to 0.01 meters.
HIGH-SIM [287]	2021	High-Granularity Highway Simulation (HIGH-SIM) consists of 2-hour 30fps vehicle trajectory dataset captured by three 8K cameras on the helicopter.	Aerial videos at multiple successive locations of Interstate-75 over 2,438m highway segment, containing data of three lanes and one off-ramp from 4:15 p.m. to 6:15 p.m.
AUTOMATUM DATA [288]	2021	AUTOMATUM Drone-based dataset includes a validation mechanism where trajectories of objects are compared against test vehicles.	Generated from 30 hours video with 18,724 vehicle miles traveled, where the relative speed error is less than 0.2 percent
MAGIC [289]	2022	Coordinated Drone video-based vehicle trajectory collected in Shanghai, China during morning peak hour	3 hours of aerial video at six points on 4,000 m of urban highway (2,000 m in each direction) with four main lanes, three off-ramps, and three on-ramps.
CitySim [290]	2022	CitySim is Drone video-based vehicle trajectory data for safety-oriented research, providing seven bounding box points for fine-grained analysis, with the capability to facilitate digital-twin-related research.	The CitySim dataset provides more than 1,200 minutes of drone video data from multiple countries in diverse weather conditions and locations.
Safety Pilot Model Deployment (SPMD) [291]	2012	The SPMD program contains Basic safety messages (BSM), vehicle trajectory, driver-vehicle interaction, and weather data are collected to evaluate V2V communication.	The SPMD data sets contain sanitized mobility data elements collected from about 3,000 vehicles, equipped with connected vehicle technologies, traversing Ann Arbor, Michigan.
Naturalistic Driving Study (NDS) [292]	2013	This dataset was collected through an extensive observational study with advanced data acquisition systems, including CAN-Bus, GPS, outward and inward facing cameras, and forward radar to capture driver behavior, vehicle movement, and environmental conditions.	More than 5 million trip files from nearly 3500 drivers located in six geographic regions across the United States.
nuScenes [293]	2020	The nuScenes dataset incorporates 6 cameras, 5 radars, and 1 LiDAR sensor. The nuScenes dataset is notable for its holistic view of detailed maps, weather conditions, and day/night cycles.	With almost 1.4 million 3D bounding boxes across 23 object classes and over 8 million annotated instances, the nuScenes dataset spans over 100 hours of driving data, capturing various urban driving conditions.
Waymo Open Dataset [294]	2020	The Waymo Open Dataset features millions of 3D bounding boxes for vehicles, pedestrians, cyclists, and other objects encountered in urban and suburban environments.	2,030 scenes for sensor data, and 103,354 scenes for motion data
Lyft Level 5 Dataset [295]	2021	This dataset captured with a fleet of autonomous vehicles navigating through urban and suburban areas over a four-month period, enhanced with high-definition maps and traffic light statuses.	This dataset, consisting of over 170,000 scenes each lasting about 25 seconds, offering detailed perceptions and motion data.
OpenACC [296]	2021	OpenACC dataset is collected for understanding of commercial ACC control logic and is designed for extensive research into vehicle performance and safety.	The dataset is a collection from 27 vehicle models across 17 brands, featuring a mix of engine types and captured using advanced GPS systems under various driving conditions.

real-world driving behaviors. PIML represents a significant step forward by embedding physical principles directly into machine learning frameworks, thereby ensuring that predictions not only fit observed data but also adhere to fundamental laws of motion. This alignment guarantees that the models remain valid under a wide range of conditions.

Another development in hybrid learning systems is Neural Symbolic AI, which seeks to bridge the gap between the intuitive pattern recognition capabilities of deep learning and the logical rigor of symbolic reasoning. By combining these two approaches, Neural Symbolic AI facilitates the construction of models that are both data-efficient and interpretable. These models excel in learning from available data while simultaneously encoding the structured knowledge that characterizes human reasoning processes. Such a dual approach enables the models to make sense of complex scenarios, predict outcomes, and make decisions that are both logically sound and grounded in real-world data. The inherent flexibility of Neural Symbolic AI allows it to adapt to new information and evolving scenarios, making it an ideal candidate for tackling the dynamic challenges of car-following behavior. As the field of autonomous driving continues to evolve, hybrid-learning systems stand poised to play a crucial role in developing models that seamlessly blend empirical data with theoretical insights, paving the way for the next generation of intelligent transportation systems that are both robust and adaptable.

World Model Predictive Control: The aspiration for driving agents to adapt swiftly to new environments and assimilate new tasks with near-human intelligence sets a high bar for the evolution of autonomous driving technologies. At the core of this evolution lies the concept of World Model Predictive Control, a paradigm shift that integrates common sense and a vast repository of background knowledge into the decision-making processes of next-generation intelligent driver models. This approach, inspired by advancements in Large Language Models (LLMs) like ChatGPT, represents a leap forward in enabling driving agents to comprehend and navigate complex driving scenarios with unprecedented depth and nuance.

The principle behind World Model Predictive Control is to endow driving agents with an internalized representation of the world. This internal model, enriched by extensive learning from diverse data sources, allows the agent to predict and plan several steps ahead, moving beyond reactive stimulus-response mechanisms to proactive situational management. Unlike traditional models that rely on immediate feedback or predefined control laws, this approach leverages predictive analytics to foresee and navigate future states, drawing on a comprehensive understanding of the driving environment's intricacies. The integration of such a world model into predictive control systems harness the full power of artificial intelligence to ensure safety and efficiency on the roads.

Hierarchical Planning and Control: The ability to understand and apply abstract concepts is a defining feature of human intelligence. These frameworks structure decision-making across multiple layers, ranging from immediate sensory responses at the lowest level to strategic planning at the highest. Incorporating hierarchical learning into intelligent driving systems facilitates the encoding of diverse factors—such as vehicle characteristics, road conditions, and traffic dynamics—into high-level representations. This allows for the formulation of complex action plans that dynamically adapt to changing conditions, thereby enhancing operational safety and predictive accuracy. The intermediate levels of this hierarchy are particularly crucial for understanding and predicting the behaviors of nearby vehicles, improving the system's ability to anticipate and react to potential hazards before they arise.

The integration of hierarchical planning and control into car-following models marks a significant step towards achieving true autonomy in driving technologies. By enabling vehicles to not only react to their immediate environment but also engage in anticipatory actions based on scenario analysis and predictive modeling, these systems enable agents to make decisions based on a blend of immediate data and abstract concepts. The development of such advanced models paves the way for a future where autonomous vehicles operate with a level of intuition and foresight akin to human drivers, contributing to safer, more efficient, and harmonious traffic ecosystems.

Multimodal Signals: The evolution of car-following models is increasingly pivoting towards the integration of multimodal signals to closely mimic the human driving experience, which inherently relies on a symphony of sensory inputs for decision-making. Current car-following models primarily utilize curated trajectory data, derived from post-processing, aggregation, and sensor fusion techniques. Human drivers base their decisions on a complex blend of visual cues, auditory signals, and even haptic feedback, something that the next generation of intelligent driving agents must emulate. By integrating information from a diverse array of sources—such as Cameras, LiDAR, Radar, Can-bus systems, Gyroscopes, and even Psychological Signals and Audio inputs—a richer, more precise understanding of the driving environment can be achieved. Approaches like multimodal fusion, which integrates data from these varied sources, cross-modality, which allows models to leverage information across different types of data, and shared representation learning, which finds commonalities across these data types, are crucial. This expanded use of multimodal signals promises to enhance the robustness, adaptability, and overall performance of car-following models, allowing them to better predict and react to the dynamic complexities of real-world driving.

Self-Supervised Learning: Traditional car-following models often rely heavily on specific objectives, predefined parameters, and extensively labeled data, which can limit their adaptability and applicability to new or unseen driving scenarios. Self-Supervised Learning (SSL) represents a paradigm shift, enabling models to learn from data that is not explicitly labeled for a specific prediction task. By exploiting the intrinsic structure of driving data, SSL techniques can uncover complex, latent relationships between past and future driving states without the need for direct supervision. This is achieved through mechanisms such as contrastive learning, which distinguishes between similar and dissimilar sequences of driving behavior, and regularized learning, which imposes additional constraints to guide the learning process. Moreover,

SSL opens up new avenues for creating models that can adapt over time, learning from continuous streams of data generated by vehicles in operation. This ongoing learning process allows models to evolve in response to changing driving conditions and behaviors, significantly enhancing their long-term utility and effectiveness. The adoption of self-supervised learning in car-following models marks a critical step towards developing autonomous driving systems that can operate with an unprecedented level of sophistication and autonomy, mirroring the complex decision-making processes of human drivers.

VII. ACKNOWLEDGEMENT

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VIII. REFERENCES

REFERENCES

[1] Reuschel A. Fahrzeugbewegungen in der Kolonne. Österreichisches Ingenieur-Archiv. 1950;4:193-215. [The Movement of a Column of Vehicles when the Leading Vehicle Is Uniformly Accelerated or Decelerated. Mag. of the Austrian Engineer and Architect Assoc.]

[2] Edie LC. Car-following and steady-state theory for noncongested traffic. *Operational Research*. 1961;9(1):66-76.

[3] Rothery RW. Car following models. Trac Flow Theory. 1992.

[4] Brackstone M, McDonald M. Car-following: a historical review. *Transportation Research Part F: Traffic Psychology and Behavior*. 1999;2(4):181-96.

[5] Olstam JJ, Tapani A. Comparison of Car-following models. Swedish National Road and Transport Research Institute; 2004. Report No.: 960.

[6] Toledo T. Driving Behavior: models and challenges. *Transport Reviews*. 2007;27(1):65-84.

[7] Wilson, R. E., & Ward, J. A. (2011). Car-following models: fifty years of linear stability analysis–a mathematical perspective. Transportation Planning and Technology, 34(1), 3-18.

[8] Li Y, Sun D. Microscopic car-following model for the Traffic flow: the state of the art. *Journal of Control Theory and Applications*. 2012;10(2):133-43.

[9] Treiber M, Kesting A. Traffic flow dynamics. *Traffic Flow Dynamics: Data, Models and Simulation*. Springer-Verlag Berlin Heidelberg; 2013. p. 983-1000.

[10] Saifuzzaman M, Zheng Z. Incorporating human-factors in car-following models: a review of recent developments and research needs. *Transportation Research Part C: Emerging Technologies*. 2014;48:379-403.

[11] van Wageningen-Kessels F, Van Lint H, Vuik K, Hoogendoorn S. Genealogy of traffic flow models. *EURO Journal on Transportation and Logistics*. 2015;4(4):445-73.

[12] Aghabayk K, Sarvi M, Young W. A state-of-the-art review of car-following models with particular considerations of heavy vehicles. Transport reviews. 2015 Jan 2;35(1):82-105.

[13] Matcha BN, Namasivayam SN, Hosseini Fouladi M, Ng KC, Sivanesan S, Eh Noum SY. Simulation strategies for mixed traffic conditions: a review of car-following models and simulation frameworks. Journal of engineering. 2020 Jan 25;2020:1-22.

[14] Ahmed HU, Huang Y, Lu P. A review of car-following models and modeling tools for human and autonomous-ready driving behaviors in micro-simulation. *Smart Cities*. 2021;4(1):314-35.

[15] Han J, Shi H, Chen L, Li H, Wang X. The car-following model and its applications in the V2X environment: A historical review. Future Internet. 2021 Dec 27;14(1):14.

[16] Han J, Wang X, Wang G. Modeling the car-following behavior with consideration of driver, vehicle, and environment factors: A historical review. Sustainability. 2022 Jul 5;14(13):8179.

[17] Sadid, H., & Antoniou, C. (2023). Modelling and simulation of (connected) autonomous vehicles longitudinal driving behavior: A state-of-the-art. IET Intelligent Transport Systems, 17(6), 1051-1071.

[18] He, Y., Zhou, Q., Wang, C., Li, J., Shuai, B., Lei, L., & Xu, H. (2023). Microscopic Modelling of Car-Following Behaviour: Developments and Future Directions. International Journal of Automotive Manufacturing and Materials, 6-6.

[19] Wang, Z., Shi, Y., Tong, W., Gu, Z., & Cheng, Q. (2023). Car-following models for human-driven vehicles and autonomous vehicles: A systematic review. Journal of transportation engineering, Part A: Systems, 149(8), 04023075.

[20] Herrey EM, Herrey H. Principles of Phys. Appl. to Traf. movements and road conditions. *American Journal of Physics*. 1945;13(1):1-14.

[21] Pipes LA. An operational Analysis of Traf. dynamics. *Journal of Applied Physics*. 1953;24(3):274-81.

[22] Kometani E, Sasaki T. On the stability of traffic flow (report-I). *Journal of the Operations Research Society of Japan*. 1958;2(1):11-26.

[23] Kometani E, Sasaki T. Dynamic behaviour of traffic with a nonlinear spacing-speed relationship. In: *Proceedings of the Symposium on Theory of Traffic Flow*, Research Laboratories, General Motors; 1959. New York: Elsevier. p. 105–119.

[24] Kometani E, Sasaki T. A safety index for traffic with linear spacing. *Operations Research*. 1959;7(6):704-20.

[25] Gipps PG. A Behavioral car-following model for Computer simulation. *Transportation Research Part B: Methodological*. 1981;15(2):105-11.

[26] Spyropoulou I. Simulation Using Gipps' Car-Following Model—An In-Depth Analysis. *Transportmetrica*. 2007;3(3):231-45.

[27] Ciuffo B, Punzo V, Montanino M. Thirty years of Gipps' car-following model: Applications, developments, and new features. *Transportation Research Record*. 2012;2315(1):89-99.

[28] Krauß S. Microscopic modeling of traffic flow: Investigation of collision free vehicle dynamics. 1998.

[29] Rakha, H., & Wang, W. Procedure for calibrating Gipps car-following model. *Transportation research record*. 2009; 2124(1), 113-124.

[30] Wilson RE. An Analysis of Gipps' Car-following Model of Highway Traffic. *IMA Journal of Applied Mathematics*. 2001;66:509-37.

[31] Shah D, Lee C, Kim YH. Modified Gipps model: a collision-free car following model. Journal of Intelligent Transportation Systems. 2023 Nov 27:1-4.

[32] Newell GF. A simplified car-following theory: a lower-order model. *Transportation Research Part B: Methodological*. 2002;36(3):195-205.

[33] Ahn S, Cassidy MJ, Laval J. Verification of a simplified car-following theory. *Transportation Research Part B: Methodological*. 2004 Jun 1;38(5):431-40.

[34] Laval, Jorge A., and Ludovic Leclercq. "A mechanism to describe the formation and propagation of stop-and-go waves in congested freeway traffic." *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*. 2010; 368, no. 1928 : 4519-4541.

[35] Chen D, Laval J, Zheng Z, Ahn S. A behavioral car-following model that captures traffic oscillations. *Transportation research part B: methodological*. 2012 Jul 1;46(6):744-61.

[36] Laval JA, Toth CS, Zhou Y. A parsimonious model for the formation of oscillations in car-following models. *Transportation Research Part B: Methodological*. 2014 Dec 1;70:228-38.

[37] Xu T, Laval J. Statistical inference for two-regime stochastic car-following models. *Transportation Research Part B: Methodological*. 2020 Apr 1;134:210-28.

[38] Taylor J, Zhou X, Rouphail NM, Porter RJ. Method for investigating intradriver heterogeneity using vehicle trajectory data: A dynamic time warping approach. *Transportation Research Part B: Methodological*. 2015 Mar 1;73:59-80.

[39] Przybyla J, Taylor J, Jupe J, Zhou X. Estimating risk effects of driving distraction: A dynamic errorable car-following model. *Transportation research part C: emerging technologies*. 2015 Jan 1;50:117-29.

[40] Meng D, Song G, Wu Y, Zhai Z, Yu L, Zhang J. Modification of Newell's car-following model incorporating multidimensional stochastic parameters for emission estimation. *Transportation Research Part D: Transport and Environment*. 2021 Feb 1;91:102692.

[41] Wolfram S. Statistical mechanics of cellular automata. *Reviews of Modern Physics*. 1983;55(3):601.

[42] Nagel K, Schreckenberg M. A cellular automaton model for freeway traffic. *Journal de physique* I. 1992;2(12):2221-29.

[43] Takayasu M, Takayasu H. 1/f noise in a traffic model. fractals. 1993 Dec;1(04):860-6.

[44] Barlovic R, Santen L, Schadschneider A, Schreckenberg M. Metastable states in cellular automata for traffic flow. *The European Physical Journal B-Condensed Matter and Complex Systems*. 1998;5:793-800.

[45] Benjamin SC, Johnson NF, Hui PM. Cellular automata models of traffic flow along a highway containing a junction. Journal of Physics A: Mathematical and General. 1996 Jun 21;29(12):3119.

[46] Knospe, W., Santen, L., Schadschneider, A. and Schreckenberg, M. (2000) Towards a realistic microscopic description of highway traffic, Journal of Physics A, 33, pp. L447–L485.

[47] Kerner BS, Klenov SL, Wolf DE. Cellular automata approach to three-phase traffic theory. *Journal of Physics A: Mathematical and General*. 2002;35(47):9971.

[48] Bham, G. H. and Benekohal, R. F. (2004) A high fidelity traffic simulation model based on cellular automata and car-following concepts, Transportation Research C, 12, pp. 1–32.

[49] Chakroborty P, Maurya AK. Microscopic analysis of cellular automata based traffic flow models and an improved model. Transport reviews. 2008 Nov 1;28(6):717-34.

[50] Daganzo CF. In Traffic flow, cellular automata = kinematic waves. *Transportation Research Part B: Methodological*. 2006;40(5):396-403.

[51] Tian J, Li G, Treiber M, Jiang R, Jia N, Ma S. Cellular automaton model simulating spatiotemporal patterns, phase transitions and concave growth pattern of oscillations in traffic flow. *Transportation Research Part B: Methodological*. 2016;93:560-75.

[52] Meng JP, Dai SQ, Dong LY, Zhang JF. Cellular automaton model for mixed traffic flow with motorcycles. Physica A: Statistical Mechanics and Its Applications. 2007 Jul 1;380:470-80.

[53] Kong D, List GF, Guo X, Wu D. Modeling vehicle car-following behavior in congested traffic conditions based on different vehicle combinations. Transportation letters. 2018 Sep 3;10(5):280-93.

[54] Mallikarjuna C, Rao KR. Cellular automata model for heterogeneous traffic. Journal of Advanced Transportation. 2009 Jul;43(3):321-45.

[55] Vasic J, Ruskin HJ. Cellular automata simulation of traffic including cars and bicycles. Physica A: Statistical Mechanics and its Applications. 2012 Apr 15;391(8):2720-9.

[56] Chen J, Li Z, Wang W, Jiang H. Evaluating bicycle–vehicle conflicts and delays on urban streets with bike lane and on-street parking. Transportation letters. 2018 Jan 2;10(1):1-1.

[57] Gavriilidou A, Daamen W, Yuan Y, Hoogendoorn SP. Modelling cyclist queue formation using a two-layer framework for operational cycling behaviour. Transportation research part C: emerging technologies. 2019 Aug 1;105:468-84.

[58] Tanveer M, Kashmiri FA, Yan H, Wang T, Lu H. A cellular automata model for heterogeneous traffic flow incorporating micro autonomous vehicles. Journal of Advanced Transportation. 2022 Jan 28;2022.

[59] Singh MK, Ramachandra Rao K. Simulation of Signalized Intersection with Non-Lane-Based Heterogeneous Traffic Conditions Using Cellular Automata. Transportation Research Record. 2023:03611981231211317.

[60] Chandler RE, Herman R, Montroll EW. Traffic dynamics: studies in car following. *Operations Research*. 1958;6(2):165-84.

[61] Herman R, Montroll EW, Potts RB, Rothery RW. Traffic dynamics: Analysis of stability in car following. *Operations Research*. 1959;7(1):86-106.

[62] Gazis DC, Herman R, Potts RB. Car-following theory of steady-state Traffic flow. *Operations Research*. 1959;7(4):499-505.

[63] Gazis DC, Herman R, Rothery RW. Nonlinear follow-the-leader models of Traffic flow. *Operations Research*. 1961;9(4):545-67.

[64] May AD, Keller HE. Non-integer car-following models. *Highway Research Record*. 1967;199(1):19-32.

[65] Heyes MP, Ashworth R. Further Research on car-following models. *Transportation Research*. 1972;6(3):287-91.

[66] Ceder A, May AD. Further evaluation of single-and two-regime Traffic flow models. *Transportation Research Record*. 1976;567.

[67] Bexelius S. An extended model for car-following. *Transportation Research*. 1968;2(1):13-21.

[68] Treiterer J, Myers J. The hysteresis phenomenon in Traffic flow. In: *Transportation and Traffic Theory*. 1974;6:13-38.

[69] Bando M, Hasebe K, Nakayama A, Shibata A, Sugiyama Y. Dynamical model of Traffic congestion and numerical simulation. *Physical Review E*. 1995;51(2):1035.

[70] Bando M, Hasebe K, Nakanishi K, Nakayama A. Analysis of optimal velocity model with explicit delay. *Physical Review E*. 1998;58(5):5429.

[71] Helbing D, Tilch B. Generalized force model of Traffic dynamics. *Physical Review E*. 1998;58(1):133.

[72] Jiang R, Wu Q, Zhu Z. Full velocity difference model for a car-following theory. *Physical Review E*. 2001;64(1):017101.

[73] Konishi K, Kokame H, Hirata K. Coupled map car-following model and its delayed-feedback Control. *Physical Review E*. 1999;60(4):4000.

[74] Lenz H, Wagner CK, Sollacher R. Multi-anticipative car-following model. *The European Physical Journal B-Condensed Matter and Complex Systems*. 1999;7(2):331-35.

[75] Nakayama A, Sugiyama Y, Hasebe K. Effect of looking at the car that follows in an optimal velocity model of Traffic flow. *Physical Review E*. 2001;65(1):016112.

[76] Hasebe K, Nakayama A, Sugiyama Y. Dynamical model of a co-operative driving System for freeway Traffic. *Physical Review E*. 2003;68(2):026102.

[77] Nagatani T. Modified KdV equation for jamming transition in the continuum models of traffic. *Physica A: Statistical Mechanics and its Applications*. 1998;261(3-4):599-607.

[78] Chiarello FA, Piccoli B, Tosin A. A statistical mechanics approach to macroscopic limits of car-following Traffic dynamics. *International Journal of Non-Linear Mechanics*. 2021;137:103806.

[79] Gong H, Liu H, Wang BH. An asymmetric full velocity difference car-following model. *Physica A: Statistical Mechanics and its Applications*. 2008;387(11):2595-2602.

[80] Jin S, Wang D, Tao P, Li P. Non-lane-based full velocity difference car following model. *Physica A: Statistical Mechanics and its Applications*. 2010;389(21):4654-4662.

[81] Yu Y, Jiang R, Qu X. A modified full velocity difference model with acceleration and deceleration confinement: calibrations, validations, and scenario analyzes. *IEEE Intelligent Transportation Systems Magazine*. 2019;13(2):222-235.

[82] Shang M, Rosenblad B, Stern R. A novel asymmetric car following model for driver-assist enabled vehicle dynamics. *IEEE Transactions on Intelligent Transportation Systems*. 2022;23(9):15696-15706.

[83] Treiber M, Hennecke A, Helbing D. Congested Traffic states in empirical observations and microscopic simulations. *Physical Review E*. 2000;62(2):1805.

[84] Treiber M, Helbing D. Memory effects in microscopic traffic models and wide scattering in flow-density data. *Physical Review E*. 2003 Oct 21;68(4):046119.

[85] Treiber M, Kesting A, Helbing D. Delays, inaccuracies and anticipation in microscopic Traffic models. *Physica A: Statistical Mechanics and its Applications*. 2006;360(1):71-88.

[86] Kesting A, Treiber M, Helbing D. Enhanced Intelligent driver model to assess the impact of driving strategies on Traffic capacity. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*. 2010;368(1928):4585-4605.

[87] Liebner M, Baumann M, Klanner F, Stiller C. Driver intent inference at urban intersections using the Intelligent driver model. In: *2012 IEEE Intelligent Vehicles Symposium*; pp. 1162-1167.

[88] Derbel O, Peter T, Zebiri H, Mourllion B, Basset M. Modified Intelligent driver model for driver safety and Traffic stability improvement. *IFAC Proceedings Volumes*. 2013;46(21):744-749.

[89] Li Z, Li W, Xu S, Qian Y. Stability Analysis of an extended Intelligent driver model and its simulations under open boundary condition. *Physica A: Statistical Mechanics and its Applications*. 2015;419:526-536.

[90] Eggert J, Damerow F, Klingelschmitt S. The foresighted driver model. In: *2015 IEEE Intelligent Vehicles Symposium*; pp. 322-329.

[91] Milanés V, Shladover SE. Modeling cooperative and autonomous adaptive cruise control dynamic responses using experimental data. *Transportation Research Part C: Emerging Technologies*. 2014;48:285-300.

[92] Hoermann S, Stumper D, Dietmayer K. Probabilistic long-term prediction for autonomous vehicles. In: *2017 IEEE Intelligent Vehicles Symposium*; 2017. pp. 237-243.

[93] Treiber M, Kesting A. The Intelligent driver model with stochasticity-new insights into Traffic flow oscillations. *Transportation Research Procedia*. 2017;23:174-187.

[94] Zhou M, Qu X, Jin S. On the impact of cooperative autonomous vehicles in improving freeway merging: a modified intelligent driver model-based approach. IEEE Transactions on Intelligent Transportation Systems. 2016 Sep 29;18(6):1422-8.

[95] Hubmann C, Schulz J, Xu G, Althoff D, Stiller C. A belief state planner for interactive merge maneuvers in congested Traffic. In: *2018 21st International Conference on Intelligent Transportation Systems (ITSC)*; 2018. pp. 1617-1624.

[96] Albeaik S, Bayen A, Chiri MT, Gong X, Hayat A, Kardous N, Keimer A, McQuade ST, Piccoli B, You Y. Limitations and improvements of the Intelligent driver model (idm). *SIAM Journal on Applied Dynamical Systems*. 2022;21(3):1862-1892.

[97] Zhang, Y., Chen, X., Wang, J., Zheng, Z., & Wu, K. (2022). A generative car-following model conditioned on driving styles. Transportation research part C: emerging technologies, 145, 103926.

[98] Zhou Z, Li L, Qu X, Ran B. A self-adaptive IDM car-following strategy considering asymptotic stability and damping characteristics. Physica A: Statistical Mechanics and its Applications. 2024 Mar 1;637:129539.

[99] Forbes TW, Simpson ME. Driver-and-Vehicle Response in Freeway Deceleration Waves. *Transportation Science*. 1968;2(1):77-104.

[100] Michaels RM, Cozan LW. Perceptual and field factors causing lateral displacement. *Highway Research Record*. 1963;25.

[101] Todosiev EP. The action point model of the driver-vehicle System. The Ohio State University; 1963.

[102] Evans L, Rothery R. Experimental measurements of perceptual thresholds in car-following. *52nd Annual Meeting of the Highway Research Board*. 1973.

[103] Wiedemann R, Reiter U. Microscopic Traffic simulation: the simulation System MISSION, background and actual state. Project ICARUS (V1052) Final Report. 1992;2:1-53.

[104] Fritzsche HT, Ag DB. A model for Traffic simulation. *Traffic Engineering & Control*. 1994;35(5):317-321.

[105] Ferrari P. The reliability of the motorway transport System. *Transportation Research Part B: Methodological*. 1988;22(4):291-310.

[106] Winsum W. The human element in car following models. *Transportation Research Part F: Traffic Psychology and Behaviour*. 1999;2:207-211.

[107] Wang WH, Zhang W, Li DH, Hirahara K, Ikeuchi K. Improved action point model in Traffic flow based on driver's cognitive mechanism. In: *Proc. of IEEE Intell. Vehicles Symp. (IV04)*; 2004; Parma, Italy. pp. 447–452.

[108] Lochrane TW, Al-Deek H, Dailey DJ, Krause C. Modeling driver behavior in work and nonwork zones: Multidimensional psychophysical car-following framework. *Transportation Research Record*. 2015;2490(1):116-126.

[109] Durrani U, Lee C. Calibration and validation of psycho-physical car-following model using driver's action points and perception thresholds. *Journal of Transportation Engineering, Part A: Systems*. 2019;145(9):04019039.

[110] Wagner P, Erdmann J, Nippold R. Action-points in human driving and in SUMO. In: *SUMO Conference Proceedings*; 2020; Vol. 1. pp. 113-122.

[111] Kikuchi S, Chakroborty P. Car-following model based on fuzzy inference System. *Transportation Research Record*. 1992;82.

[112] Chakroborty P, Kikuchi S. Evaluation of the General Motors based car-following models and a proposed fuzzy inference model. *Transportation Research Part C: Emerging Technologies*. 1999;7(4):209-235.

[113] Zheng P, McDonald M. Application of fuzzy System in the car-following Behavior Analysis. In: *International Conference on Fuzzy Systems and Knowledge Discovery*; 2005. Springer, Berlin, Heidelberg. pp. 782-791.

[114] Chai C, Wong YD. Fuzzy cellular automata model for signalized intersections. *Computer-Aided Civil and Infrastructure Engineering*. 2015 Dec;30(12):951-64.

[115] Hao H, Ma W, Xu H. A fuzzy logic-based multi-agent car-following model. *Transportation Research Part C: Emerging Technologies*. 2016;69:477-496.

[116] Bennajeh A, Bechikh S, Said LB, Aknine S. A fuzzy logic-based anticipation car-following model. In: *Transactions on Computational Collective Intelligence*; Springer, Cham; 2018. pp. 200-222.

[117] Hoberock LL, Rouse RJ Jr. Emergency Control of vehicle platoons: System operation and platoon leader Control. *Journal of Dynamic Systems, Measurement, and Control*. 1976;98(3):245-251.

[118] Chiu HY, Stupp GB Jr., Brown SJ Jr. Vehicle-follower Control with variable gains for short headway automated guideway transit System. *Journal of Dynamic Systems, Measurement, and Control*. 1977;99(3):183-189.

[119] Shladover SE. Longitudinal Control of automated guideway transit vehicles within platoons. *Journal of Dynamic Systems, Measurement, and Control*. 1978;100:302-310.

[120] Cho D, Hedrick JK. Automotive powertrain modelling for Control. *Journal of Dynamic Systems, Measurement, and Control*. 1989;111.

[121] McMahon DH, Hedrick JK, Shladover SE. Vehicle modeling and Control for automated highway Systems. In: *Proceedings of the 1990 American Control Conference*; 1990; San Diego, CA. pp. 297-303.

[122] Suzuki H, Nakatsuji T. Effect of adaptive cruise Control (ACC) on Traffic throughput: numerical example on actual freeway corridor. *JSAE review*. 2003;24(4):403-410.

[123] Ploeg J, Serrarens AF, Heijenk GJ. Connect & Drive: design and evaluation of cooperative adaptive cruise Control for congestion reduction. *Journal of Modern Transportation*. 2011;19(3):207-213.

[124] Milanés V, Shladover SE, Spring J, Nowakowski C, Kawazoe H, Nakamura M. Cooperative adaptive cruise control in real traffic situations. IEEE Transactions on intelligent transportation systems. 2013 Aug 30;15(1):296-305.

[125] Shladover SE, Nowakowski C, Lu XY, Ferlis R. Cooperative Adaptive Cruise Control: Definitions and Operating Concepts. *Transportation Research Record*. 2015;2489(1):145-152.

[126] Swaroop DV, Hedrick JK, Chien CC, Ioannou P. A comparison of spacing and headway Control laws for Automatically Controlled vehicles. *Vehicle System Dynamics*. 1994;23(1):597-625.

[127] Li PY, Shrivastava A. Traffic flow stability induced by constant time headway policy for adaptive cruise Control vehicles. *Transportation Research Part C: Emerging Technologies*. 2002;10(4):275-301.

[128] Rakha, H., Snare, M., & Dion, F. (2004). Vehicle dynamics model for estimating maximum light-duty vehicle acceleration levels. *Transportation Research Record*, 1883(1), 40-49.

[129] Rakha, H.A.; Ahn, K.; Faris, W.; et al. Simple vehicle powertrain model for modeling intelligent vehicle applications.*IEEE Transactions on Intelligent Transportation Systems*. 2012,13(2),770–780.

[130] Fadhloun, K.; Rakha, H.; Loulizi, A. Vehicle dynamics model for estimating typical vehicle accelerations. *Transportation Research Record*. 2015,2491(1),61–71.53.

[131] Fadhloun, K.; Rakha, H. A novel vehicle dynamics and human behavior car-following model: Model development and preliminary testing. *International Journal of Transportation Science and Technology*. 2020,9(1),14–28.

[132] Helly W. Simulation of bottlenecks in single lane traffic flow. In: *Theory of Traffic Flow* proceedings; 1959. Amsterdam: Elsevier.

[133] Milanés V, Villagrá J, Godoy J, González C. Comparing fuzzy and Intelligent PI Controllers in stop-and-go manoeuvres. *IEEE Transactions on Control Systems Technology*. 2012;20:770-778.

[134] Kim SG, Tomizuka M, Cheng KH. Smooth motion Control of the adaptive cruise Control System by a virtual lead vehicle. *International Journal of Automotive Technology*. 2012;13(1):77-85.

[135] Dias JEA, Pereira GAS, Palhares RM. Longitudinal model identification and velocity Control of an autonomous car. *IEEE Transactions on Intelligent Transportation Systems*. 2014;16(2):776-786.

[136] Jin IG, Orosz G. Optimal Control of connected vehicle System with communication delay and driver reaction time. *IEEE Transactions on Intelligent Transportation Systems*. 2016;18(8):2056-2070.

[137] Wang, Yanbing. "RECONSTRUCTION OF MIXED TRAFFIC SYSTEMS AT MICRO AND MACRO SCALES." PhD diss., 2023.

[138] VanderWerf J, Shladover S, Kourjanskaia N, Miller M, Krishnan H. Modeling effects of driver Control assistance Systems on Traffic. *Transportation Research Record*. 2001;1748(1):167-174.

[139] Holve R, Protzel P, Bernasch J, Naab K. Adaptive fuzzy Control for driver assistance in car-following. In: *Proceedings of the 3rd EUFIT Conference*; 1995. pp. 1149-1153.

[140] Naranjo JE, González C, Reviejo J, García R, De Pedro T. Adaptive fuzzy Control for inter-vehicle gap keeping. *IEEE Transactions on Intelligent Transportation Systems*. 2003;4(3):132-142.

[141] Pérez J, Milanés V, Godoy J, Villagra J, Onieva E. Cooperative Controllers for highways based on human experience. *Expert Systems with Applications*. 2013;40(4):1024-1033.

[142] Zhang L, Orosz G. Motif-based design for connected vehicle System in presence of heterogeneous connectivity structures and time delays. *IEEE Transactions on Intelligent Transportation Systems*. 2016;17(6):1638-1651.

[143] Azizıaghdam ET, Alankuş OB. Longitudinal Control of Autonomous Vehicles Consisting Powertrain With Nonlinear Characteristics. *IEEE Transactions on Intelligent Vehicles*. 2022;7(1):133-142.

[144] Wang Y, Wang Z, Han K, Tiwari P, Work DB. Gaussian process-based personalized adaptive cruise control. *IEEE Transactions on Intelligent Transportation Systems*. 2022;23(11):21178-21189.

[145] Bageshwar VL, Garrard WL, Rajamani R. Model predictive Control of transitional maneuvers for adaptive cruise Control vehicles. *IEEE Transactions on Vehicular Technology*. 2004;53(5):1573-1585.

[146] Moser D, Waschl H, Kirchsteiger H, Schmied R, Del Re L. Cooperative adaptive cruise Control applying stochastic linear model predictive Control strategies. In: *2015 European Control Conference (ECC)*; 2015. pp. 3383-3388.

[147] Hoogendoorn S, Hoogendoorn R, Wang M, Daamen W. Modeling driver, driver support, and cooperative Systems with dynamic optimal Control. *Transportation Research Record*. 2012;2316(1):20-30.

[148] Wang M, Treiber M, Daamen W, Hoogendoorn SP, van Arem B. Modelling supported driving as an optimal Control cycle: Framework and model characteristics. *Procedia - Social and Behavioral Sciences*. 2013;80:491-511.

[149] Zhou Y, Ahn S, Chitturi M, Noyce DA. Rolling horizon stochastic optimal control strategy for ACC and CACC under uncertainty. Transportation Research Part C: Emerging Technologies. 2017 Oct 1;83:61-76.

[150] Cheng S, Li L, Mei MM, Nie YL, Zhao L. Multiple-objective adaptive cruise Control System integrated with DYC. *IEEE Transactions on Vehicular Technology*. 2019;68(5):4550-4559.

[151] Zhou Y, Wang M, Ahn S. Distributed model predictive control approach for cooperative car-following with guaranteed local and string stability. Transportation research part B: methodological. 2019 Oct 1;128:69-86.

[152] Kehtarnavaz N, Groswold N, Miller K, Lascoe P. A transportable neural-network approach to autonomous vehicle following. *IEEE Transactions on Vehicular Technology*. 1998;47(2):694-702.

[153] Panwai S, Dia H. Neural agent car-following models. *IEEE Transactions on Intelligent Transportation Systems*. 2007;8(1):60-70.

[154] Khodayari A, Ghaffari A, Kazemi R, Braunstingl R. A modified car-following model based on a neural network model of the human driver effects. *IEEE Transactions on Systems, Man, and Cybernetics - Part A: Systems and Humans*. 2012;42(6):1440-1449.

[155] Zheng J, Suzuki K, Fujita M. Car-following behavior with instantaneous driver–vehicle reaction delay: A neural-network-based methodology. *Transportation Research Part C: Emerging Technologies*. 2013;36:339-351.

[156] Wei D, Liu H. Analysis of asymmetric driving behavior using a self-learning approach. *Transportation Research Part B: Methodological*. 2013;47:1-14.

[157] Papathanasopoulou V, Antoniou C. Towards data-driven car-following models. *Transportation Research Part C: Emerging Technologies.* 2015 Jun 1;55:496-509.

[158] Dabiri, S., & Abbas, M. (2018). Evaluation of the Gradient Boosting of Regression Trees Method on Estimating Car-Following Behavior. Transportation Research Record, 2672(45), 136-146.

[159] Yang, D., Zhu, L., Liu, Y., Wu, D., & Ran, B. A novel car-following Control model combining Machine learning and kinematics models for automated vehicles. *IEEE Transactions on Intelligent Transportation Systems*. 2018.

[160] Soldevila, I. E., Knoop, V. L., & Hoogendoorn, S. (2021). Car-Following Described by Blending Data-Driven and Analytical Models: A Gaussian Process Regression Approach. Transportation Research Record, 2675(12), 1202-1213.

[161] Kamjoo E, Saedi R, Zockaie A, Ghamami M, Gates T, Talebpour A. Developing Car-Following Models for Winter Maintenance Operations Incorporating Machine Learning Methods. *Transportation Research Record*. 2022.

[162] Morton J, Wheeler TA, Kochenderfer MJ. Analysis of recurrent neural networks for probabilistic modeling of driver behavior. *IEEE Transactions on Intelligent Transportation Systems*. 2016 Sep 13;18(5):1289-98.

[163] Zhou M, Qu X, Li X. A recurrent neural network based microscopic car following model to predict Traffic oscillation. *Transportation Research Part C: Emerging Technologies*. 2017;84:245-264.

[164] Wang X, Jiang R, Li L, Lin Y, Zheng X, Wang FY. Capturing car-following behaviors by deep learning. *IEEE Transactions on Intelligent Transportation Systems*. 2017;19(3):910-920.

[165] Wang, X., Jiang, R., Li, L., Lin, Y. L., & Wang, F. Y. (2019). Long memory is important: A test study on deep-learning based car-following model. *Physica A: Statistical Mechanics and its Applications*, 514, 786-795.

[166] Wu Y, Tan H, Chen X, Ran B. Memory, attention and prediction: a deep learning architecture for car-following. *Transportmetrica B: Transport Dynamics*. 2019;7(1):1553-1571.

[167] Zhang X, Sun J, Qi X, Sun J. Simultaneous modeling of car-following and lane-changing behaviors using deep learning. *Transportation Research Part C: Emerging Technologies*. 2019;104:287-304.

[168] Ma L, Qu S. A sequence to sequence learning based car-following model for multistep predictions considering reaction delay. *Transportation Research Part C: Emerging Technologies*. 2020;120:102785.

[169] Zhou J, Wan J, Zhu F. Transfer Learning Based Long Short-Term Memory Car-Following Model for Adaptive Cruise Control. *IEEE Transactions on Intelligent Transportation Systems*. 2022.

[170] Xu D, Gao G, Qiu Q, Li H. A car-following model considering missing data based on TransGAN networks. *IEEE Transactions on Intelligent Vehicles*. 2023.

[171] Lu, W., Yi, Z., Liang, B., Rui, Y., & Ran, B. (2023). Learning Car-Following Behaviors for a Connected Automated Vehicle System: An Improved Sequence-to-Sequence Deep Learning Model. IEEE Access, 11, 28076-28089.

[172] Xu, N., Chen, C., Zhang, Y., Wang, J., Liu, Q., & Guo, C. (2024). A Sequence-to-Sequence Car-Following Model for Addressing Driver Reaction Delay and Cumulative Error in Multi-Step Prediction. IEEE Transactions on Intelligent Transportation Systems.

[173] Chen, X., Yuan, X., Zhu, M., Zheng, X., Shen, S., Wang, X., & Wang, F. Y. (2024). AggFollower: Aggressiveness Informed Car-Following Modeling. IEEE Transactions on Intelligent Vehicles.

[174] Ng AY, Russell S. Algorithms for inverse reinforcement learning. In *ICML*; 2000; Vol. 1. p. 2.

[175] Shimosaka M, Kaneko T, Nishi K. Modeling risk anticipation and defensive driving on residential roads with inverse reinforcement learning. In *17th International IEEE Conference on Intelligent Transportation Systems*; 2014. IEEE. pp. 1694-1700.

[176] Shimosaka M, Nishi K, Sato J, Kataoka H. Predicting driving behavior using inverse reinforcement learning with multiple reward functions towards environmental diversity. In *2015 IEEE Intelligent Vehicles Symposium*; 2015. IEEE. pp. 567-572.

[177] Kuefler A, Morton J, Wheeler T, Kochenderfer M. Imitating driver behavior with generative adversarial networks. In: *2017 IEEE Intelligent Vehicles Symposium*; 2017. pp. 204-211. IEEE.

[178] Gao H, Shi G, Xie G, Cheng B. Car-following method based on inverse reinforcement learning for autonomous vehicle decision-making. *International Journal of Advanced Robotic Systems*. 2018;15(6):1729881418817162.

[179] Zhou Y, Fu R, Wang C. Learning the car-following behavior of drivers using maximum entropy deep inverse reinforcement learning. *Journal of Advanced Transportation*. 2020.

[180] Zhao Z, Wang Z, Han K, Gupta R, Tiwari P, Wu G, Barth MJ. Personalized car following for autonomous driving with inverse reinforcement learning. In *2022 International Conference on Robotics and Automation* (ICRA); 2022. IEEE. pp. 2891-2897.

[181] Koutsopoulos HN, Farah H. Latent class model for car following behavior. *Transportation Research Part B: methodological*. 2012;46(5):563-578.

[182] He Z, Zheng L, Guan W. A simple nonparametric car-following model driven by field data. *Transportation Research Part B: Methodological*. 2015;80:185-201.

[183] Lefevre S, Carvalho A, Borrelli F. A learning-based framework for velocity control in autonomous driving. *IEEE Transactions on Automation Science and Engineering*. 2015;13(1):32-42.

[184] Guha A, Lei R, Zhu J, Nguyen X, Zhao D. Robust unsupervised learning of temporal dynamic vehicle-to-vehicle interactions. *Transportation research part C: emerging technologies*. 2022;142:103768.

[185] Ioannu PA, Chien CC. Autonomous Intelligent Cruise Control. *IEEE Transactions on Vehicular Technology*. 1993;42(4):657-672.

[186] Desjardins C, Chaib-Draa B. Cooperative adaptive cruise Control: A reinforcement learning approach. *IEEE Transactions on Intelligent Transportation Systems*. 2011;12(4):1248-1260.

[187] Zhu M, Wang X, Wang Y. Human-like autonomous car-following model with deep reinforcement learning. *Transportation Research Part C: Emerging Technologies*. 2018;97:348-368.

[188] Zhang Y, Guo L, Gao B, Qu T, Chen H. Deterministic promotion reinforcement learning Applied to longitudinal velocity Control for automated vehicles. *IEEE Transactions on Vehicular Technology*. 2019;69(1):338-348.

[189] Li M, Li Z, Xu C, Liu T. Deep reinforcement learning-based vehicle driving strategy to reduce crash risks in Traffic oscillations. *Transportation Research Record*. 2020;2674(10):42-54.

[190] Yen YT, Chou JJ, Shih CS, Chen CW, Tsung PK. Proactive car-following using deep-reinforcement learning. In *2020 IEEE 23rd International Conference on Intelligent Transportation Systems*; 2020. IEEE. pp. 1-6.

[191] Masmoudi M, Friji H, Ghazzai H, Massoud Y. A reinforcement learning framework for video frame-based autonomous car-following. *IEEE Open Journal of Intelligent Transportation Systems*. 2021;2:111-127.

[192] Yavas U, Kumbasar T, Ure NK. Model-Based Reinforcement Learning for Advanced Adaptive Cruise Control: A Hybrid Car Following Policy. In *2022 IEEE Intelligent Vehicles Symposium*; 2022. IEEE. pp. 1466-1472.

[193] Wang Z, Huang H, Tang J, Meng X, Hu L. Velocity Control in car-following behavior with autonomous vehicles using reinforcement learning. *Accident Analysis & Prevention*. 2022;174:106729.

[194] Tang X, Chen J, Yang K, Toyoda M, Liu T, Hu X. Visual Detection and Deep Reinforcement Learning-Based Car Following and Energy Management for Hybrid Electric Vehicles. *IEEE Transactions on Transportation Electrification*. 2022;8(2):2501-2515.

[195] Liao, Y., Yu, G., Chen, P., Zhou, B., & Li, H. (2024). Modelling personalised car-following behaviour: a memory-based deep reinforcement learning approach. Transportmetrica A: transport science, 20(1), 2035846.

[196] Cui, Y., Huang, S., Zhong, J., Liu, Z., Wang, Y., Sun, C., ... & Khajepour, A. (2023). DriveLLM: Charting the path toward full autonomous driving with large language models. IEEE Transactions on Intelligent Vehicles.

[197] Chen, L., Sinavski, O., Hünermann, J., Karnsund, A., Willmott, A. J., Birch, D., ... & Shotton, J. (2023). Driving with llms: Fusing object-level vector modality for explainable autonomous driving. arXiv preprint arXiv:2310.01957.

[198] Wen, L., Fu, D., Li, X., Cai, X., Ma, T., Cai, P., ... & Qiao, Y. (2023). Dilu: A knowledge-driven approach to autonomous driving with large language models. arXiv preprint arXiv:2309.16292.

[199] Cui, C., Ma, Y., Cao, X., Ye, W., & Wang, Z. (2024). Drive as you speak: Enabling human-like interaction with large language models in autonomous vehicles. In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (pp. 902-909).

[200] Wayve. (2023). LINGO-1: Exploring natural language for autonomous driving. Retrieved from https://wayve.ai/thinking/lingo-natural-language-autonomous-driving/

[201] Wayve. (2023). LINGO-2: Driving with language. Retrieved from https://wayve.ai/thinking/lingo-2-driving-with-language/

[202] Tian, X., Gu, J., Li, B., Liu, Y., Hu, C., Wang, Y., ... & Zhao, H. (2024). DriveVLM: The Convergence of Autonomous Driving and Large Vision-Language Models. arXiv preprint arXiv:2402.12289.

[203] Cui, C., Ma, Y., Cao, X., Ye, W., Zhou, Y., Liang, K., ... & Zheng, C. (2024). A survey on multimodal large language models for autonomous driving. In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (pp. 958-979).

[204] Wang, W., Xie, J., Hu, C., Zou, H., Fan, J., Tong, W., ... & Dai, J. (2023). Drivemlm: Aligning multi-modal large language models with behavioral planning states for autonomous driving. arXiv preprint arXiv:2312.09245.

[205] Shao, H., Hu, Y., Wang, L., Waslander, S. L., Liu, Y., & Li, H. (2023). Lmdrive: Closed-loop end-to-end driving with large language models. arXiv preprint arXiv:2312.07488.

[206] ResearchRabbit. (2023). ResearchRabbit: Your Research Supercharged. Retrieved from https://www.researchrabbit.ai

[207] Li, Y., Lu, X., Ren, C., & Zhao, H. (2019). Fusion modeling method of car-following characteristics. *IEEE Access*, 7, 162778-162785.

[208] Mo Z, Shi R, Di X. A Physics-informed deep learning paradigm for car-following models. *Transportation Research Part C: Emerging Technologies*. 2021;130:103240.

[209] Shiomi, Y., Li, G., & Knoop, V. L. (2023). Analysis of stochasticity and heterogeneity of car-following behavior based on data-driven modeling. Transportation research record, 2677(12), 604-619.

[210] Pan, W., Zhang, J., Tian, J., Cui, F., & Wang, T. (2023). Analysis of car–following behaviors based on data–driven and theory–driven car–following models: Heterogeneity and asymmetry. Physica A: Statistical Mechanics and its Applications, 632, 129324.

[211] Chen D, Srivastava A, Ahn S, Li T. Traffic dynamics under speed disturbance in mixed traffic with automated and non-automated vehicles. *Transportation research part C: emerging technologies*. 2020;113:293-313.

[212] Kontar W, Li T, Srivastava A, Zhou Y, Chen D, Ahn S. On multi-class automated vehicles: Car-following behavior and its implications for traffic dynamics.*Transportation research part C: emerging technologies*. 2021;

[213] Ye L, Yamamoto T. Modeling connected and autonomous vehicles in heterogeneous traffic flow. *Physica A: Statistical Mechanics and its Applications*. 2018;490:269-277.

[214] Li Y, Zhang L, Zheng H, He X, Peeta S, Zheng T, Li Y. Evaluating the energy consumption of electric vehicles based on car-following model under non-lane discipline. *Nonlinear Dynamics*. 2015;82:629-641.

[215] Zhu WX, Zhang HM. Analysis of mixed traffic flow with human-driving and autonomous cars based on car-following model. *Physica A: Statistical Mechanics and its Applications*. 2018;496:274-285.

[216] Talebpour A, Mahmassani HS. Influence of connected and autonomous vehicles on traffic flow stability and throughput. *Transportation research part C: emerging technologies*. 2016;71:143-163.

[217] Rahman MS, Abdel-Aty M. Longitudinal safety evaluation of connected vehicles' platooning on expressways. *Accident Analysis & Prevention*. 2018;117:381-391.

[218] Huang Y, Ye Y, Sun J, Tian Y. Characterizing the Impact of Autonomous Vehicles on Macroscopic Fundamental Diagrams. *IEEE Transactions on Intelligent Transportation Systems*. 2023.

[219] Sun J, Zheng Z, Sun J. Stability analysis methods and their applicability to car-following models in conventional and connected environments. *Transportation research part B: methodological*. 2018;109:212-237.

[220] An S, Xu L, Qian L, Chen G, Luo H, Li F. Car-following model for autonomous vehicles and mixed traffic flow analysis based on discrete following interval. *Physica A: Statistical Mechanics and its Applications*. 2020;560:125246.

[221] Sun D, Chen D, Zhao M, Liu W, Zheng L. Linear stability and nonlinear analyses of traffic waves for the general nonlinear car-following model with multi-time delays. *Physica A: Statistical Mechanics and its Applications*. 2018;501:293-307.

[222] Ma, H., Wu, H., Hu, Y., Chen, Z., & Luo, J. (2022). Multiple-factors aware car-following model for connected and autonomous vehicles. Transportation research record, 2676(2), 649-662.

[223] Kamjoo E, Saedi R, Zockaie A, Ghamami M, Gates T, Talebpour A. Developing Car-Following Models for Winter Maintenance Operations Incorporating Machine Learning Methods. *Transportation Research Record*. 2023;2677(2):519-540.

[224] Garg M, Bouroche M. Can connected autonomous vehicles improve mixed traffic safety without compromising efficiency in realistic scenarios?. IEEE Transactions on Intelligent Transportation Systems. 2023 Jan 26.

[225] M. Krishnamurthy and C. -T. Huang, "Safety Assessment of CACC Vehicles with Faulty Sensors Using SUMO Simulations," SoutheastCon 2022, Mobile, AL, USA, 2022, pp. 58-63, doi: 10.1109/SoutheastCon48659.2022.9764090.

[226] Fleming J, Yan X, Lot R. Incorporating driver preferences into eco-driving assistance systems using optimal control. IEEE Transactions on Intelligent Transportation Systems. 2020 Mar 9;22(5):2913-22.

[227] Li SE, Xu S, Huang X, Cheng B, Peng H. Eco-departure of connected vehicles with V2X communication at signalized intersections. IEEE Transactions on Vehicular Technology. 2015 Sep 29;64(12):5439-49.

[228] S. E. Li et al., "Performance Enhanced Predictive Control for Adaptive Cruise Control System Considering Road Elevation Information," in IEEE Transactions on Intelligent Vehicles, vol. 2, no. 3, pp. 150-160, Sept. 2017, doi: 10.1109/TIV.2017.2736246.

[229] Xin Q, Fu R, Yuan W, Liu Q, Yu S. Predictive intelligent driver model for eco-driving using upcoming traffic signal information. Physica A: Statistical Mechanics and its Applications. 2018 Oct 15;508:806-23.

[230] Rawashdeh ZY, Mahmud SM. A novel algorithm to form stable clusters in vehicular ad hoc networks on highways. Eurasip journal on wireless communications and networking. 2012 Dec;2012:1-3.

[231] Sun Z, Huang T, Zhang P. Cooperative decision-making for mixed traffic: A ramp merging example. *Transportation research part C: emerging technologies*. 2020;120:102764.

[232] Li T, Hui F, Zhao X, Liu C, Ngoduy D. Modelling heterogeneous traffic dynamics by considering the influence of V2V safety messages. *IET Intelligent Transport Systems*. 2020;14(4):220-227.

[233] Jia D, Ngoduy D. Platoon based cooperative driving model with consideration of realistic inter-vehicle communication. *Transportation Research Part C: Emerging Technologies*. 2016;68:245-264.

[234] Li T, Han X, Ma J. Cooperative perception for estimating and predicting microscopic traffic states to manage connected and automated traffic. *IEEE Transactions on Intelligent Transportation Systems*. 2021;23(8):13694-13707.

[235] Navas F, Milanés V. Mixing V2V-and non-V2V-equipped vehicles in car following. *Transportation research part C: emerging technologies*. 2019;108:167-181.

[236] Meng Y, Li L, Wang FY, Li K, Li Z. Analysis of cooperative driving strategies for nonsignalized intersections. *IEEE Transactions on Vehicular Technology*. 2017;67(4):2900-2911.

[237] Jia D, Ngoduy D. Enhanced cooperative car-following traffic model with the combination of V2V and V2I communication. *Transportation Research Part B: Methodological*. 2016;90:172-191.

[238] Taniguchi Y, Nishi R, Ezaki T, Nishinari K. Jam-absorption driving with a car-following model. *Phys. A, Statist. Mech. Appl.* 2015;433:304–315.

[239] Nishi R. Theoretical conditions for restricting secondary jams in jam-absorption driving scenarios. *Physica A: Statistical Mechanics and its Applications*. 2020;542:123393.

[240] Kesting A, Treiber M, Schönhof M, Helbing D. Adaptive cruise control design for active congestion avoidance. *Transportation Research Part C: Emerging Technologies*. 2008;16(6):668-683.

[241] Stern RE, Cui S, Delle Monache ML, Bhadani R, Bunting M, Churchill M, Hamilton N, Pohlmann H, Wu F, Piccoli B, Seibold B. Dissipation of stop-and-go waves via control of autonomous vehicles: Field experiments. *Transportation Research Part C: Emerging Technologies*. 2018;89:205-221.

[242] Cui S, Seibold B, Stern R, Work DB. Stabilizing traffic flow via a single autonomous vehicle: Possibilities and limitations. In *2017 IEEE Intelligent Vehicles Symposium (IV)*; 2017; Los Angeles, CA, USA. IEEE. pp. 1336-1341.

[243] Chen T, Wang M, Gong S, Zhou Y, Ran B. Connected and automated vehicle distributed control for on-ramp merging scenario: A virtual rotation approach. *Transportation Research Part C: Emerging Technologies*. 2021;133:103451.

[244] Wang Y, Jin PJ. Model predictive control policy design, solutions, and stability analysis for longitudinal vehicle control considering shockwave damping. *Transportation Research Part C: Emerging Technologies*. 2023;148:104038.

[245] Vinitsky E, Kreidieh A, Le Flem L, Kheterpal N, Jang K, Wu C, Wu F, Liaw R, Liang E, Bayen AM. Benchmarks for reinforcement learning in mixed-autonomy traffic. in *Conference on Robot Learning*, pp. 399-409, 2018.

[246] Fu Z, Kreidieh AR, Wang H, Lee JW, Delle Monache ML, Bayen AM. Cooperative Driving for Speed Harmonization in Mixed-Traffic Environments. In*2023 IEEE Intelligent Vehicles Symposium (IV)* 2023 Jun 4 (pp. 1-8). IEEE.

[247] Hayat A, Piccoli B, Truong S. Dissipation of traffic jams using a single autonomous vehicle on a ring road. SIAM Journal on Applied Mathematics. 2023 Jun 30;83(3):909-37.

[248] Hu X, Wang H, Tang X. Cyber-physical control for energy-saving vehicle following with connectivity. IEEE Transactions on industrial electronics. 2017 May 11;64(11):8578-87.

[249] Jin S, Sun DH, Zhao M, Li Y, Chen J. Modeling and stability analysis of mixed traffic with conventional and connected automated vehicles from cyber physical perspective. Physica A: Statistical Mechanics and its Applications. 2020 Aug 1;551:124217.

[250] Wang P, Wu X, He X. Modeling and analyzing cyberattack effects on connected automated vehicular platoons. *Transportation Research Part C: Emerging Technologies*. 2020;115:102625.

[251] Wang P, Yu G, Wu X, Wang Y, He X. Spreading patterns of malicious information on single-lane platooned traffic in a connected environment. *Computer-Aided Civil and Infrastructure Engineering*. 2019;34(3):248-265.

[252] Nice M, Elmadani S, Bhadani R, Bunting M, Sprinkle J, Work D. CAN coach: vehicular control through human cyber-physical systems. InProceedings of the ACM/IEEE 12th International Conference on Cyber-Physical Systems 2021 May 19 (pp. 132-142).

[253] Wang Z, Han K, Tiwari P. Digital twin-assisted cooperative driving at non-signalized intersections. *IEEE Transactions on Intelligent Vehicles*. 2021;7(2):198-209.

[254] McQuade ST, Denaro C, Mahmood M, Lee JW, Gumm G, Sprinkle JM, Work DB, Piccoli B, Seibold B, Bayen AM. Medium-scale to large-scale implementation of cyber-physical human experiments in live traffic. IFAC-PapersOnLine. 2022 Jan 1;55(41):83-8.

[255] Li H, Jin S. Intelligent vehicle car-following model based on cyber physical system and its simulation under mixed traffic flow. Physica A: Statistical Mechanics and its Applications. 2024 Jan 15;634:129482.

[256] Raveendran, B., Mathew, T. V., & Velaga, N. R. (2024). Modeling car-following behavior during queue discharge at signalized intersections with countdown timer. Transportation letters, 16(1), 1-13.

[257] Saifuzzaman M, Zheng Z, Haque MM, Washington S. Revisiting the Task–Capability Interface model for incorporating human factors into car-following models. *Transportation Research Part B: Methodological*. 2015;82:1-19.

[258] Chen X, Sun J, Ma Z, Sun J, Zheng Z. Investigating the long-and short-term driving characteristics and incorporating them into car-following models. *Transportation Research Part C: Emerging Technologies*. 2020;117:102698.

[259] Ro JW, Roop PS, Malik A, Ranjitkar P. A formal approach for modeling and simulation of human car-following behavior. *IEEE Transactions on Intelligent Transportation Systems*. 2017;19(2):639-648.

[260] Van Lint JWC, Calvert SC. A generic multi-level framework for microscopic traffic simulation—Theory and an example case in modeling driver distraction. *Transportation Research Part B: Methodological*. 2018;117:63-86.

[261] Harth M, Bin Amjad U, Kates R, Bogenberger K. Incorporation of human factors to a data-driven car-following model. *Transportation Research Record*. 2022;2676(10):291-302.

[262] Huang X, Sun J, Sun J. A car-following model considering asymmetric driving behavior based on long short-term memory neural networks. *Transportation Research Part C: Emerging Technologies*. 2018;95:346-362.

[263] Hoel CJ, Driggs-Campbell K, Wolff K, Laine L, Kochenderfer MJ. Combining planning and deep reinforcement learning in tactical decision making for autonomous driving. *IEEE Transactions on Intelligent Vehicles*. 2019;5(2):294-305.

[264] Sazara C, Nezafat RV, Cetin M. Offline reconstruction of missing vehicle trajectory data from 3D LIDAR. In *2017 IEEE Intelligent Vehicles Symposium (IV)*; 2017 Jun 11. IEEE. pp. 792-797.

[265] Montanino M, Punzo V. Trajectory data reconstruction and simulation-based validation against macroscopic traffic patterns. *Transportation Research Part B: Methodological*. 2015 Oct 1;80:82-106.

[266] Chen X, Yin J, Qin G, Tang K, Wang Y, Sun J. Integrated macro-micro modelling for individual vehicle trajectory reconstruction using fixed and mobile sensor data. *Transportation Research Part C: Emerging Technologies*. 2022 Dec 1;145:103929.

[267] Chen X, Yin J, Tang K, Tian Y, Sun J. Vehicle trajectory reconstruction at signalized intersections under connected and automated vehicle environment. *IEEE Transactions on Intelligent Transportation Systems*. 2022 Feb 17;23(10):17986-8000.

[268] Wang Y, Wei L, Chen P. Trajectory reconstruction for freeway traffic mixed with human-driven vehicles and connected and automated vehicles. *Transportation Research Part C: Emerging Technologies*. 2020 Feb 1;111:135-55.

[269] Wei L, Wang Y, Chen P. A particle filter-based approach for vehicle trajectory reconstruction using sparse probe data. *IEEE Transactions on Intelligent Transportation Systems*. 2020 Mar 9;22(5):2878-90.

[270] Lin Y, Wang P, Zhou Y, Ding F, Wang C, Tan H. Platoon trajectories generation: A unidirectional interconnected LSTM-based car-following model. *IEEE Transactions on Intelligent Transportation Systems*. 2020 Dec 1;23(3):2071-81.

[271] Arman MA, Tampère CM. Lane-level trajectory reconstruction based on data-fusion. *Transportation Research Part C: Emerging Technologies*. 2022 Dec 1;145:103906.

[272] Makridis MA, Kouvelas A. Adaptive physics-informed trajectory reconstruction exploiting driver behavior and car dynamics. *Scientific Reports*. 2023 Jan 20;13(1):1121.

[273] Shan X, Hao P, Chen X, Boriboonsomsin K, Wu G, Barth MJ. Vehicle energy/emissions estimation based on vehicle trajectory reconstruction using sparse mobile sensor data. *IEEE Transactions on Intelligent Transportation Systems*. 2018 May 11;20(2):716-26.

[274] Feng Y, Sun J, Chen P. Vehicle trajectory reconstruction using automatic vehicle identification and traffic count data. *Journal of Advanced Transportation*. 2015 Mar;49(2):174-94.

[275] Zhao J, Yang X, Zhang C. Vehicle trajectory reconstruction for intersections: an integrated wavelet transform and Savitzky-Golay filter approach. *Transportmetrica A: Transport Science*. 2023 Jan 3.

[276] Chen X, Zhu M, Chen K, Wang P, Lu H, Zhong H, Han X, Wang X, Wang Y. FollowNet: a comprehensive benchmark for car-following behavior modeling. Scientific data. 2023 Nov 25;10(1):828.

[277] Zhang TT, Jin PJ, Piccoli B, Sartipi M. Deep spatial-temporal embedding for vehicle trajectory validation and refinement. Computer-Aided Civil and Infrastructure Engineering. 2024 Jan 30.

[278] G. Dahiya, Y. Asakura and W. Nakanishi, "A study of speed-density functional relations for varying spatiotemporal resolution using Zen Traffic Data," 2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC), Rhodes, Greece, 2020, pp. 1-8, doi: 10.1109/ITSC45102.2020.9294564.

[279] Krajewski R, Bock J, Kloeker L, Eckstein L. The highd dataset: A drone dataset of naturalistic vehicle trajectories on german highways for validation of highly automated driving systems. In2018 21st international conference on intelligent transportation systems (ITSC) 2018 Nov 4 (pp. 2118-2125). IEEE.

[280] Zyner A, Worrall S, Nebot EM. Acfr five roundabouts dataset: Naturalistic driving at unsignalized intersections. IEEE Intelligent Transportation Systems Magazine. 2019 Sep 11;11(4):8-18.

[281] Zhan W, Sun L, Wang D, Shi H, Clausse A, Naumann M, Kummerle J, Konigshof H, Stiller C, de La Fortelle A, Tomizuka M. Interaction

dataset: An international, adversarial and cooperative motion dataset in interactive driving scenarios with semantic maps. arXiv preprint arXiv:1910.03088. 2019 Sep 30.

[282] Bock J, Krajewski R, Moers T, Runde S, Vater L, Eckstein L. The ind dataset: A drone dataset of naturalistic road user trajectories at german intersections. In2020 IEEE Intelligent Vehicles Symposium (IV) 2020 Oct 19 (pp. 1929-1934). IEEE.

[283] Krajewski R, Moers T, Bock J, Vater L, Eckstein L. The round dataset: A drone dataset of road user trajectories at roundabouts in germany. In2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC) 2020 Sep 20 (pp. 1-6). IEEE.

[284] Gloudemans D, Wang Y, Ji J, Zachar G, Barbour W, Hall E, Cebelak M, Smith L, Work DB. I-24 MOTION: An instrument for freeway traffic science. Transportation Research Part C: Emerging Technologies. 2023 Oct 1;155:104311.

[285] Barmpounakis E, Geroliminis N. On the new era of urban traffic monitoring with massive drone data: The pNEUMA large-scale field experiment. Transportation research part C: emerging technologies. 2020 Feb 1;111:50-71.

[286] UTE database, Ubiquitous Traffic Eyes (UTE), 2021. On-line. Available: http://seutraffic.com

[287] Shi X, Zhao D, Yao H, Li X, Hale DK, Ghiasi A. Video-based trajectory extraction with deep learning for High-Granularity Highway Simulation (HIGH-SIM). Communications in transportation research. 2021 Dec 1;1:100014.

[288] Spannaus P, Zechel P, Lenz K. Automatum data: Drone-based highway dataset for the development and validation of automated driving software for research and commercial applications. In2021 IEEE Intelligent Vehicles Symposium (IV) 2021 Jul 11 (pp. 1372-1377). IEEE.

[289] Ma W, Zhong H, Wang L, Jiang L, Abdel-Aty M. MAGIC dataset: multiple conditions unmanned aerial vehicle group-based high-fidelity comprehensive vehicle trajectory dataset. Transportation research record. 2022 May;2676(5):793-805.

[290] Zheng, O., Abdel-Aty, M., Yue, L., Abdelraouf, A., Wang, Z., & Mahmoud, N. (2023). CitySim: A Drone-Based Vehicle Trajectory Dataset for Safety-Oriented Research and Digital Twins. Transportation Research Record, 0(0).

[291] Huang X, Zhao D, Peng H. Empirical study of DSRC performance based on safety pilot model deployment data. IEEE Transactions on Intelligent Transportation Systems. 2017 Jan 31;18(10):2619-28.

[292] Blatt A, Pierowicz J, Flanigan M, Lin PS, Kourtellis A, Lee C, Jovanis P, Jenness J, Wilaby M, Campbell J, Richard C. Naturalistic driving study: Field data collection. 2015.

[293] Caesar H, Bankiti V, Lang AH, Vora S, Liong VE, Xu Q, Krishnan A, Pan Y, Baldan G, Beijbom O. nuscenes: A multimodal dataset for autonomous driving. InProceedings of the IEEE/CVF conference on computer vision and pattern recognition 2020 (pp. 11621-11631).

[294] Sun P, Kretzschmar H, Dotiwalla X, Chouard A, Patnaik V, Tsui P, Guo J, Zhou Y, Chai Y, Caine B, Vasudevan V. Scalability in perception for autonomous driving: Waymo open dataset. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition 2020 (pp. 2446-2454).

[295] Li G, Jiao Y, Knoop VL, Calvert SC, van Lint JW. Large Car-following Data Based on Lyft level-5 Open Dataset: Following Autonomous Vehicles vs. Human-driven Vehicles. arXiv preprint arXiv:2305.18921. 2023 May 30.

[296] Makridis M, Mattas K, Anesiadou A, Ciuffo B. OpenACC. An open database of car-following experiments to study the properties of commercial ACC systems. Transportation research part C: emerging technologies. 2021 Apr 1;125:103047.